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***Predicting Daily Changes in ICIS TTF
Front-Month Prices with Machine
Learning***

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Abstract

European gas prices have been highly volatile over the past 5 years with disruptive impacts on people, energy systems and transition trajectories across the world. The Dutch Title Transfer Facility (TTF) sets the reference price for the continent, but very little academic literature exists using modern data-driven methods to analyse the market. This dissertation aims to bridge this gap by applying machine learning models to predict future daily changes in TTF prices, testing for informational market inefficiencies and quantifying which variables drive price change during these periods. The findings show that no model tested can consistently predict future daily changes over long periods at a statistically significant level, but certain models can do so during specific periods, with the XGBoost model standing out as the best performing model. Key drivers identified included, during the 2021-2023 energy crisis, news articles related to geopolitical events, and in the post-crisis period, variables reflecting past industrial gas demand and future space heating demand. These findings contribute to academic analysis of the TTF market, provide energy market participants with realistic evidence on the forecasting value of machine learning models, and can help policymakers better understand the behaviour of a market which is critical to Europe's energy security and ongoing energy transition.

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1. Introduction and research aim

Natural gas plays a central role in the European energy system, serving as a critical input for electricity generation, industrial production and residential heating. The Dutch Title Transfer Facility (TTF) has emerged as the principal hub for pricing natural gas in Europe, reflected in its deep market liquidity relative to other European trading hubs (Heather, 2025). Over time, the TTF has also gained global prominence, increasingly serving as a benchmark for liquefied natural gas (LNG) in regions beyond Europe (Heather, 2025).

The importance of TTF gas prices was starkly highlighted during the 2021-2023 energy crisis, which was driven initially by a rapid post-pandemic economic rebound and then sharply intensified by the Russian invasion of Ukraine, which cut off a significant portion of pipeline gas supply from Europe. In response, TTF gas prices surged to record highs, sending shockwaves through the energy system. Not only did natural gas itself become significantly more expensive, but electricity prices also rose sharply, reflecting the role of gas-fired power plants as marginal generators in many European markets (Zakeri et al., 2023). The resulting increases in energy bills exacerbated inflationary pressures (Adolfson, 2024) and helped to fuel a continent-wide cost of living crisis.

The effects of soaring TTF prices were felt globally as LNG cargoes were redirected towards Europe, leaving cash-strapped lower-income countries relying on LNG in the cold. This led to widespread rolling blackouts in countries such as Pakistan (Stapczynski and Mangi, 2023) and Bangladesh (Paul & Varadhan, 2022), increasing energy poverty and social conflicts (Zhang and Wang, 2024).

These disruptions reveal the extent to which TTF price movements have become globally consequential, influencing energy security, economic stability, and energy transition trajectories in import-dependent nations. For these reasons, it is highly important to understand what shapes TTF prices. This research contributes to that effort, in addition to deepening the literature on price prediction in energy markets and application of machine learning models.

The aim of the research is to evaluate whether machine learning models can predict future day-on-day changes in ICIS TTF M+1 prices using supply/demand variables. In the process, the informational efficiency of the TTF market will be assessed and variables with particular importance identified. This is done by training five models on a dataset spanning 1 January 2016 – 31 May 2025 and testing to see whether any of the models can outperform a naïve no-change

forecast at a statistically significant level, either across entire dataset or within a period of the dataset.

2. Literature review

This project sits at the intersection of three adjacent fields of research: 1. Analysis of the European gas market, 2. Application of machine learning algorithms, 3. Price prediction in financial and energy markets. The following literature review is divided into a separate section for each, with the aim to highlight major developments and relevant research, as well as motivate the project by identifying research gaps. Section 1 describes the literature on gas price determinants, which will inform feature selection, Section 2 describes key themes in modern machine learning, which will shape model discussion, and Section 3 discusses price prediction and the efficient market hypothesis.

2.1 What drives the price of natural gas?

Natural gas prices are shaped by a complex interplay of supply-side, demand-side, and exogenous factors. These include traditional fundamentals such as storage levels, production volumes, and weather, as well as more structural and emergent influences such as geopolitical risk, financialisation of energy markets, and global LNG trade dynamics. In contrast to the relatively mature literature on oil pricing, research on natural gas price formation remains relatively underdeveloped, with much of it focusing on US natural gas markets rather than on European markets.

In the U.S. context, Brown (2008) used an error-correction model on data between 1997-2007 to find a long-run relationship between natural gas prices and crude oil prices, with additional impact from weather and gas inventories. Roberts (2019) replicated this analysis, with the same result, but also finding that when the sample was expanded to include data up to the 2017, the long-run relationship between crude oil and natural gas prices had completely disappeared. Roberts postulated that this could be a result of the increase in U.S. natural gas supply associated with the shale revolution beginning in the mid-2000s. Also drawing on econometric methods, Hailemariam and Smyth (2019) applied a structural heterogeneous autoregressive VAR model to find that demand shocks, such as unexpected surges in residential or industrial demand, were the primary drivers of U.S. natural gas price movements between 1978-2018.

In the European context, Nick and Thoenes (2014) developed a structural autoregressive VAR model for the German gas market, finding that German natural gas prices between 2008-2012 were primarily affected by temperature, storage and supply shortfalls in the short term, while

the long-term development was tied to both crude oil and coal prices. Hulshof et al., (2016) obtained similar results for the TTF market, using TTF Day-ahead prices between 2011-2014, finding that weather and storage availability were the key price determinants, with a small impact from oil prices and no effect from coal prices.

In addition to the role of fundamentals and prices of substitutable commodities, the rapid rise in financial players active on the TTF in recent years has brought attention to the role of speculators in Europe's natural gas market, with market observers noting a strong correlation between speculator positioning and TTF prices. There has been limited recent work on this topic, but Obadi and Matej (2023) found, using Granger causality tests, that speculator's market positioning was driven by fundamental factors between January 2019 – June 2022, and was not a main factor driving the price of natural gas.

Most of the existing literature on natural gas price determinants has relied on traditional econometric approaches, using data from before the 2021-2023 energy crisis. This provides a research gap which this project will contribute to filling by using data up to 31 May 2025 and applying machine learning models to identify key drivers.

2.2 From classical statistics to modern machine learning

The modern field of machine learning has its roots in classical statistics. Galton's regression to the mean (1886) and Pearson's correlation coefficient (1896) provided early tools for modelling linear relationships, while Fisher's maximum-likelihood framework (1922) formalised estimation and inference. By mid-century, statisticians confronted the curse of dimensionality; Tikhonov's ridge regularisation (1943) introduced penalty terms to prevent overfitting, a theme central to nearly all modern algorithms. Decision theory (von Neumann & Morgenstern, 1944) formalised optimisation under uncertainty, and Shannon's information theory (1948) introduced entropy, which underlies the splitting criteria of decision trees.

The first recognisably "machine learning" algorithm was Rosenblatt's perceptron (1958). With the development of backpropagation (Rumelhart et al., 1986) and hardware accelerators such as GPUs, neural networks grew into deep learning, producing architectures such as LSTMs (Hochreiter & Schmidhuber, 1997), CNNs (Krizhevsky et al., 2012) and Transformers (Vaswani et al., 2017).

Ensemble tree-based methods, combining multiple decision trees, matured in parallel. Significant developments included Breiman's Random Forests (2001) which reduce prediction variance by averaging predictions across trees trained on bootstrapped samples and random

subsets of features, and gradient boosting methods such as XGBoost (Chen & Guestrin, 2016), which improve accuracy by building trees sequentially so that each one corrects the errors of the previous. These approaches incorporate built-in regularisation through sampling and averaging and have been found to outperform deep learning approaches on tabular data with less than 10,000 samples (Grinsztajn, 2022).

The dataset used in this research is tabular in structure and contains ~2,000 samples, suggesting the tree-based models may outperform the deep learning architectures. To evaluate this hypothesis, the study compares the predictive performance of two ensemble methods - Random Forest and XGBoost – against two neural networks architectures - a standard feedforward artificial neural network and a long short-term memory (LSTM) network - with linear regression serving as a baseline benchmark. In doing so, the study contributes to the growing body of evidence on the relative performance of machine learning models for structured, medium-sized datasets in energy market forecasting.

2.3 Price forecasting in financial and energy markets

Financial and energy market analysts have strived to apply breakthroughs in statistics and machine learning over the past decades to forecast the future value of financial assets. A notable early approach was the vector-autoregressive (VAR) model, introduced by Sims (1980), which became particularly influential in macroeconomics and commodity price modelling. VAR models are typically used for modelling interconnected variables, such as interest rates, inflation and energy prices, and are based on the assumption that past values of one variable can help forecast another.

However, the predictive ambitions of early financial models were tempered by Fama's (1970) Efficient Market Hypothesis (EMH), which argues that asset prices fully reflect all available information. Under the EMH, systemic price prediction becomes theoretically impossible, as any exploitable pattern based on previous values would instantly be arbitrated away. This implies that past prices cannot help to predict future prices and that asset prices behave like a random walk (Samuelson, 1965). While the EMH has been challenged, especially in the context of short-term anomalies, it continues to pose a central constraint on predictive modelling in liquid, competitive markets (Malkiel, 2003).

In the context of energy markets, efforts to predict prices have been most successful in electricity markets, largely due to favourable physical characteristics, market structure and data availability. Electricity markets typically operate through centrally organised day-ahead

auctions where publicly available demand forecasts must be met by generators submitting bids to supply electricity. As electricity cannot be stored economically at large scale, supply and demand must match in real time, creating predictable patterns in prices such as seasonality and autocorrelation, which violate EMH and can be exploited by machine learning models to consistently outperform naïve baselines at a statistically significant level (Weron, 2014; Jędrzejewski, 2022; Lago et al., 2021).

In contrast, oil and gas markets are decentralised and continuously traded, often via bilateral contracts or over-the-counter (OTC) deals, lacking real-time transparency for key fundamentals such as storage and supply outages. This makes oil and gas markets more opaque and asynchronous, reducing the quality and timeliness of inputs available to predictive models (Nick & Thones, 2014). Furthermore, unlike electricity, oil and gas are storable on a large scale, making prices less regionally and temporally contained and more exposed to geopolitical events with potential to disrupt supply over time (El-Gamal & Jaffe, 2018). This combination of lagged and fragmented data, storability, and a highly liquid but globally exposed trading environment makes oil and gas prices inherently noisy and difficult to forecast.

A review on oil price forecasting (Ellwanger & Snudden, 2023) found that existing forecasts do not outperform naïve last-observed-value forecasts for predictions up to one year ahead. A review by Conlon et al., (2022) came to the same conclusion regarding the predictability of monthly crude oil spot returns.

As for natural gas price forecasting, the existing literature has primarily been focused on the U.S. market. Čeperić et al., (2017) compared the performance of neural networks and classical time series models for U.S. Henry Hub gas prices, finding that without careful preselection of variables, the machine learning models performed worse than naïve baselines such as a random walk model or historical average model. They noted that “machine learning results in literature often over exaggerate their successfulness.” Also looking at the U.S. market, Mouchtaris et al., (2021) tested machine learning methods on 2423 samples of Henry Hub spot prices with 21 external variables, finding that bagged trees narrowly outperformed the random walk at prediction horizons 1, 3, 5 and 10 days ahead, however the statistical significance of these results were not evaluated. More recently, Baumeister et al., (2024) used a Bayesian VAR model to forecast Henry Hub spot prices, finding that a naïve last-observed-value predictor gives the best forecast for prediction horizons up to 1 month.

In Europe, recent literature on natural gas price prediction with machine learning remains scarce. Focusing on the German market, which is significantly less liquid than the Dutch TTF,

Böhm et al., (2023) used artificial neural networks (ANNs) for price prediction, finding that the ANNs outperformed simple naïve predictors. Similarly, Aziz and Simensen (2023) forecasted German gas prices using a temporal fusion transformer, reporting reductions in model error against benchmark models.

Focusing on the TTF market, Sørensen (2023) found that LSTMs outperformed traditional econometric models such as Generalised Autoregressive Conditional Heteroskedasticity (GARCH) models in predicting price volatility. On the other hand, Baltas et al., (2025) found that GARCH models outperformed neural networks in long-term six-month forecasts of TTF prices.

As far as the author is aware, the only existing study forecasting daily TTF M+1 prices is by Berrisch and Ziel (2022), who built probabilistic forecasting models for day-ahead and month-ahead products, reporting reductions in out-of-sample continuous ranked probability score (CPRS) compared to benchmark models. This provides a clear research gap and motivation for this research, which will contribute evidence on the performance of machine learning point forecasts of daily changes in TTF M+1 prices.

3. Methodology

3.1 Learning objectives and research workflow

The learning objective for this research is to investigate whether daily changes in TTF M+1 prices can be predicted using machine learning models. The available TTF M+1 time series includes prices on UK working days in the period 1 January 2016 – 31 May 2025, consisting of 2375 observations.

Instead of predicting the absolute price for each day, the models are tested on predicting how much the ICIS settlement price will change in €/MWh from one day to the next. This is done to avoid scaling issues, as absolute prices varied widely during the studied period, and to reflect that market participants in practice will be more concerned with changes in price rather than the absolute level.

The models are tested using two distinct approaches:

1. **Look-back:** In this approach, the model is only given data from D-1 or before to predict the price change from D-1 to day D. Results are compared to a naïve baseline which predicts a price change of 0 from D-1 to D. If any models tested perform significantly better than the naïve baseline, this would suggest a violation of the efficient market

hypothesis for the TTF, since it would mean that information from the past can be used to predict future prices.

2. **Look-forward:** In this approach, the model is given the same explanatory variables as under the look-back approach, but it also gets to peak into the next 14 days average space heating demand, gas-for-power demand and LNG flows. If models under the look-forward approach perform significantly better than the naïve baseline, it suggests that accurate weather forecasts and ship tracking services can help predict TTF M+1 gas prices.

The look-forward approach is motivated by the fact that in practice, market participants have access to reasonably accurate 14-day weather forecasts which are the primary drivers of space heating demand and power demand. Furthermore, global LNG arrivals can typically be reasonably well deduced two weeks in advance by using ship tracking services, as the journeys from global LNG supply centres (US Gulf Coast & Qatar) to global LNG demand centres (Northwest Europe & East Asia) typically take at least 14 days.

Under the two approaches, models are tested on both the entire dataset and on 6 subsets of the dataset to compare model performance over time and to reflect that pricing regimes and the relative importance of fundamental factors can change from one period to the next. The 6 subsets are obtained by splitting the original TTF M+1 timeseries into 6 distinct periods using a binary segmentation algorithm from the Python library *ruptures* (Truong, Oudre, & Vayatis, 2020).

This algorithm detects changes in mean and variance and is specified to include at least 100 points for each period. Below is the result of the change points algorithm segmentation.

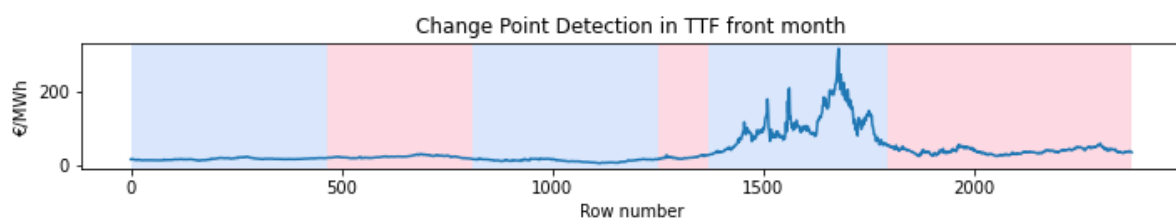


Figure 1: Result of change point detection algorithm segmentation

Period 5 stands out as a very high-price/high-volatility period, spanning from 4 June 2021 to 9 February 2023, representing the 2021-2023 energy crisis. It is also worth noting that Period 4, representing the months leading up to the energy crisis, has a particularly low number of

observations compared to the other periods. Below is a breakdown of key metrics for each period.

	Date from	Date to	Mean €/MWh	Variance	Rows
Period 1	1 January 2016	1 November 2017	15.26	5.76	465
Period 2	2 November 2017	14 March 2019	21.55	9.23	345
Period 3	15 March 2019	7 December 2020	11.02	12.48	440
Period 4	8 December 2020	3 June 2021	19.76	9.95	120
Period 5	4 June 2021	9 February 2023	104.61	2901.53	425
Period 6	10 February 2023	31 May 2025	37.48	56.24	580

Table 1: Description of each period in period split

For each period, a model is trained on all but the last 60 rows and tested on the last 60 rows, corresponding to predictions spanning roughly 12 working weeks per period. In addition to training and testing on each period, the models are also evaluated against an expanding window and a sliding window, which are designed to mimic model deployment in a real-world trading environment.

In the expanding window, the training and testing set is sequentially updated so that the training data gradually grows. In the first iteration, the model is trained on the first 300 rows and tested on the next 30. The second iteration trains on the first 330 rows, testing on the next 30 rows and so on. By the end of the walk-forward validation, all but the first 300 rows have been tested on all the past data.

The sliding window is identical except the training set doesn't grow in size, instead the train and test windows are kept fixed at 300/30 and slid forward until the end of the dataset.

For both the expanding and sliding windows, evaluation metrics are aggregated across all the testing periods to compare against the naïve baseline. If any model performs significantly better than the naïve baseline in either of the walk-forward approaches, it suggests a violation of informational efficiency for the TTF market not just in an isolated period but across the entire 2016-2025 period.

The flowchart on the following page describes the overall research workflow.

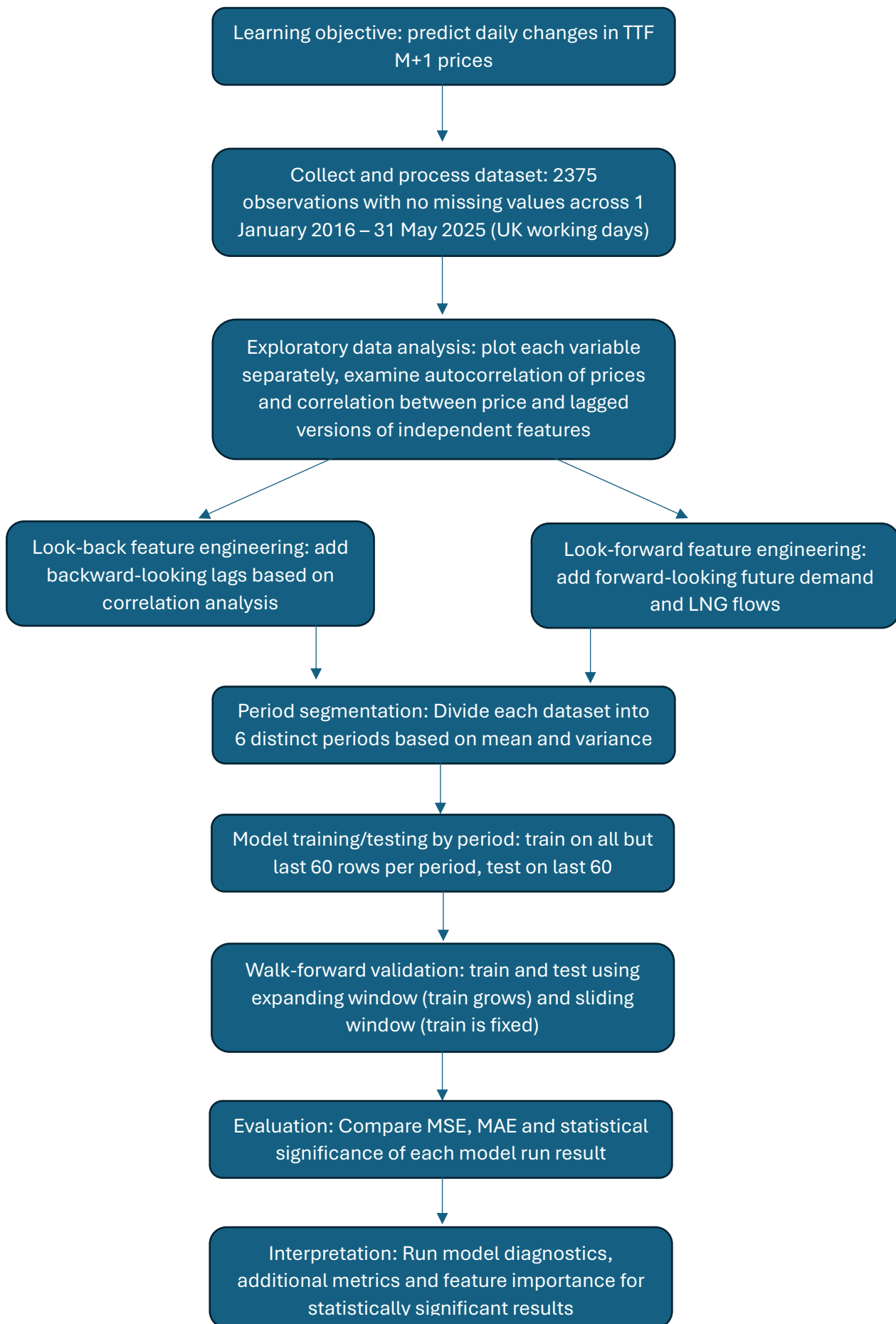


Figure 2: Flowchart describing research workflow

The remaining sections in the methodology will describe in more detail the data collected, feature engineering performed, evaluation metrics used, and model architectures employed.

3.2 Data description

Data variables collected for this study were carefully selected based on domain knowledge and the emphasis found in the literature on key drivers of gas prices – namely, demand, storage, LNG flows and geopolitical risk. Early project experimentation found that including more variables introduced too much noise to the detriment of model performance – therefore the feature set was deliberately restricted to the most relevant variables in order to reduce model overfitting.

Pricing data, demand data, storage data, and LNG flow data were obtained from ICIS, a leading provider of energy market data. The Geopolitical Risk (GPR) Index was sourced from Caldara and Iacoviello (2022). The following section provides a description of each variable included in the dataset and plots them to them to inspect outliers and confirm data quality.

3.2.1 Prices

The prices to be predicted in this study are benchmark ICIS TTF front-month contract prices. The front-month contract, or M+1, is the most liquid TTF contract and typically sets the price for the rest of Europe. ICIS publishes independent benchmark price assessments for the TTF M+1 which are widely referenced by the industry and used for hedging purposes. The ICIS TTF M+1 is assessed at 16:30 London time on UK working days (ICIS, 2024) and expires at the end of each month. Upon expiry, the seller is required to deliver gas in equal agreed quantities across each day of the next month, although in practice many market participants close or roll their positions before expiry to avoid physical delivery.

The price graph below shows TTF M+1 prices over the analysis period, displaying a strong upswing in volatility and absolute price level during the 2021-2023 energy crisis.

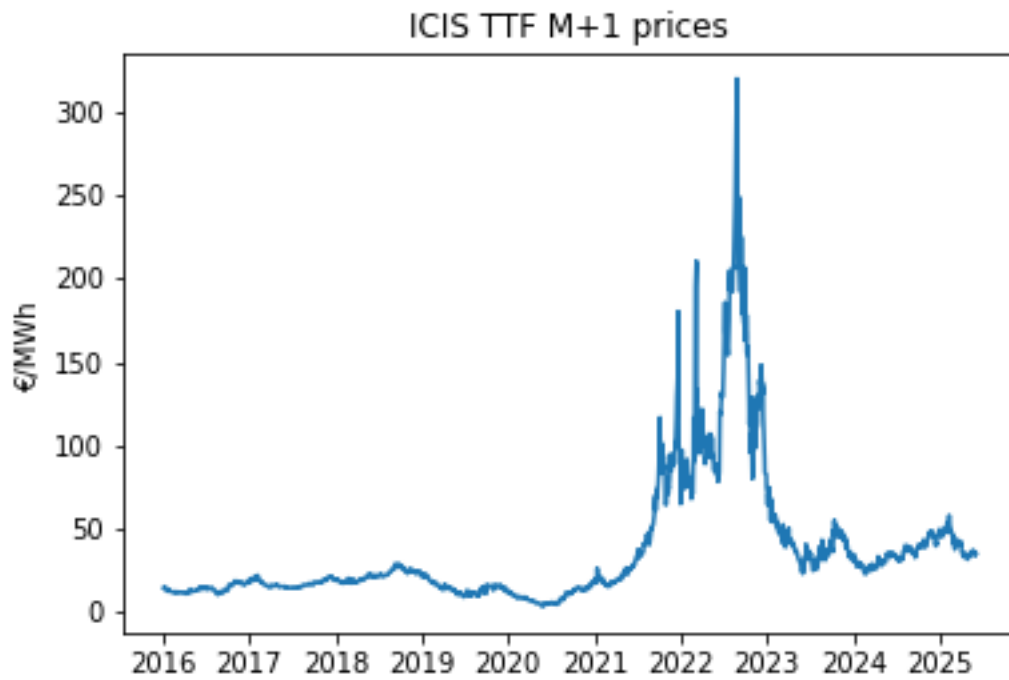


Figure 3: Plot of ICIS TTF M+1 settlement prices over analysis period

As discussed in methodology section 3.1, rather than predicting absolute prices, the models will be predicting daily price changes, as shown below.

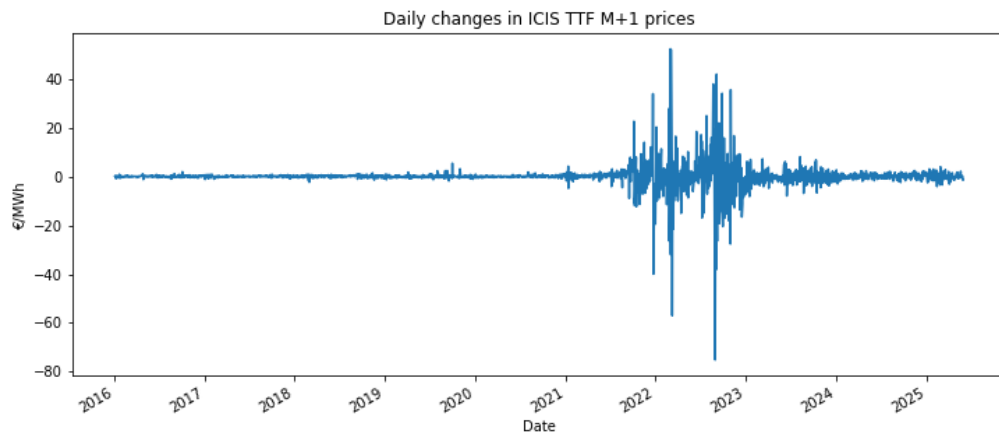


Figure 4: Plot of daily changes in ICIS TTF M+1 prices over analysis period

3.2.2 Demand

European gas demand can be divided into demand from three sectors: space heating in the residential and commercial sector, power production, and industrial production. Since TTF prices serve as a proxy for European-wide gas prices, the demand data used was aggregated on

a European level with data for the following 12 countries: Great Britain, Netherlands, Germany, France, Belgium, Austria, Italy, Spain, Czech Republic, Slovakia, Poland and Hungary.

The sector-level demand dataset has a daily granularity and is only available from 1 January 2016, which is why total dataset is limited to analysing the period between 1 January 2016 – 31 May 2025. As the graph below shows, overall gas demand is highly cyclical, driven by primarily by strong peaking demand for space heating during the winter.

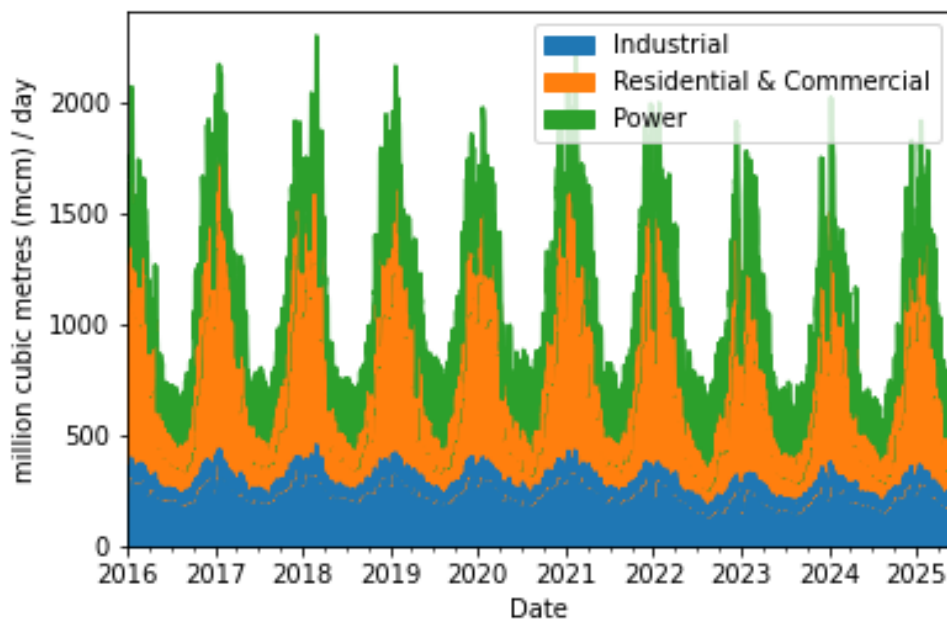


Figure 5: Stacked area plot of aggregated European gas demand by sector

3.2.3 Storage

The storage variable used in the project quantifies how much of European storage is filled on any given day and takes a value from 0 to 1, where 1 represents that 100% of storage capacity is full. The data is aggregated by ICIS based on European storage operators and covers the following countries: Austria, Belgium, Bulgaria, Croatia, Czech Republic, Denmark, France, Germany, Great Britain, Hungary, Italy, Latvia, Netherlands, Poland, Portugal, Romania, Slovakia, Spain and Sweden. Strategic stocks are included. Notably, Ukrainian storage is not included.

As the graph below shows, European gas storage levels are seasonally cyclical, with gas typically injected into storage during the summer and withdrawn from storage during the winter. Storage levels were unusually low from 2021 to mid-2022 due to the cut in Russian supply.

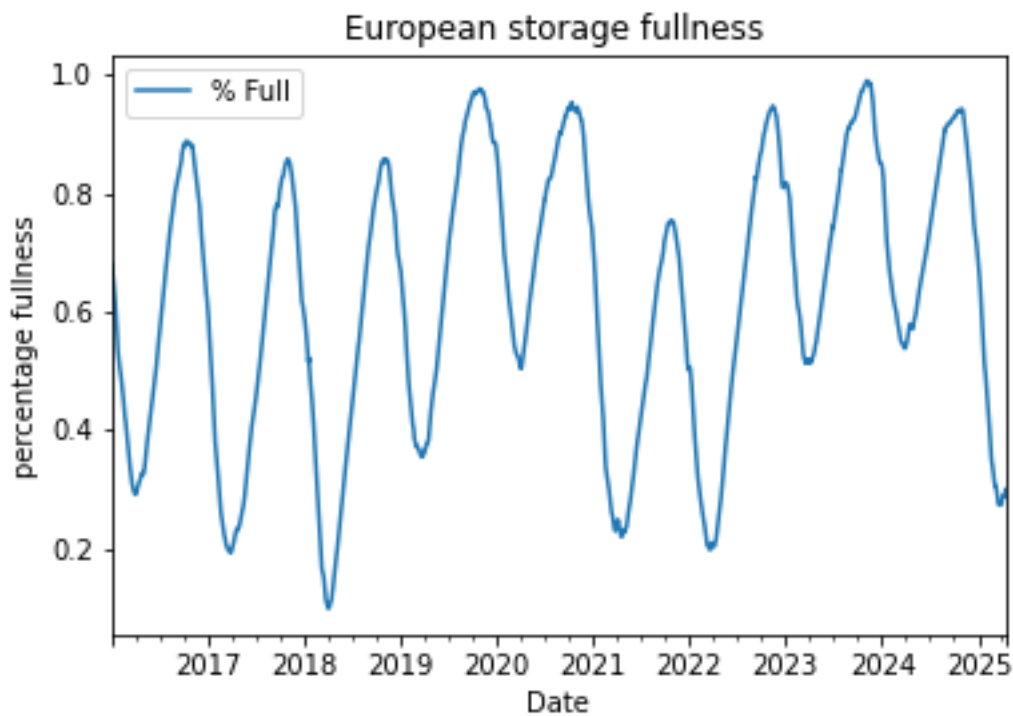


Figure 6: Plot of aggregate European storage levels, excluding Ukraine

3.2.4 LNG flows

LNG flow data was collected from the ICIS LNG Edge platform, which tracks global LNG cargo voyages. This data was processed to produce a count for each day of LNG cargo arrivals in Europe, Asia, and elsewhere.

LNG arrival destinations classed as 'Europe' were: Belgium, Croatia, Finland, France, Germany, United Kingdom, Greece, Italy, Lithuania, Netherlands, Poland, Portugal and Spain.

Destinations classed as 'Asia' were: Japan, China, Taiwan, South Korea, India, Pakistan, Thailand, Indonesia, Vietnam and the Philippines.

All other destinations were classed as 'Other'.

As graph below shows, there was sharp increase in cargo arrivals to Europe between 2022-2023, as Europe had to replace Russian pipeline gas with LNG deliveries.

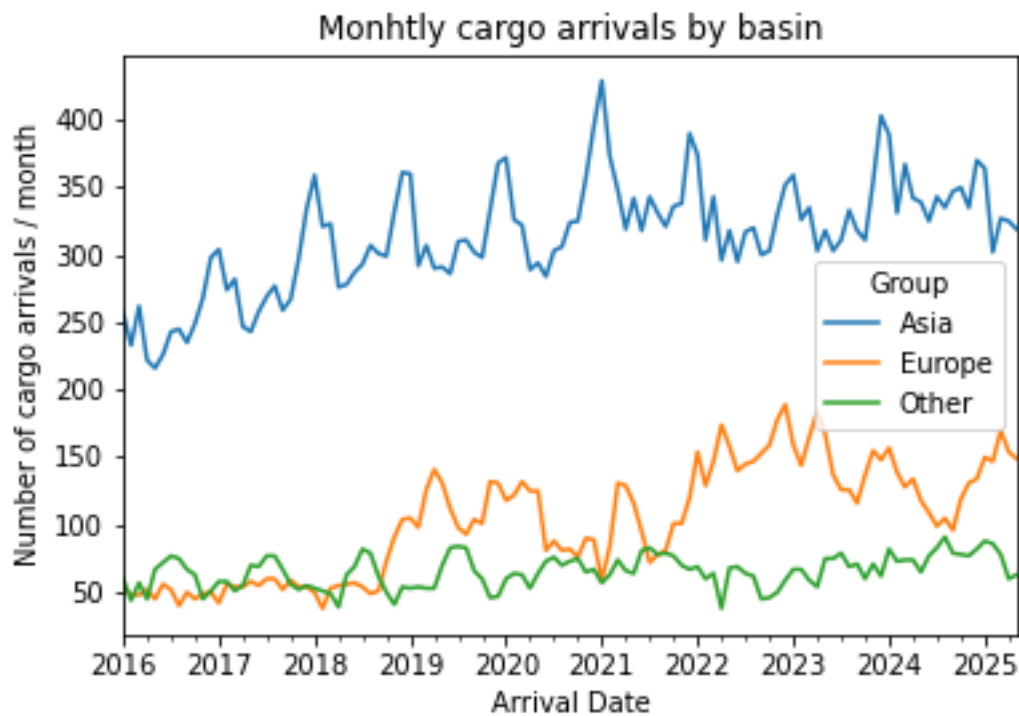


Figure 7: Plot of monthly cargo arrivals to Europe, Asia and elsewhere

3.2.5 Geopolitical risk

The geopolitical risk indicator was sourced from Caldara and Iacoviello (2022) and counts daily occurrences of articles related to adverse geopolitical events from 10 newspapers: Chicago Tribune, the Daily Telegraph, Financial Times, The Globe and Mail, The Guardian, the Los Angeles Times, The New York Times, USA Today, The Wall Street Journal, and The Washington Post.

This variable serves as a proxy for military conflicts and political tensions which could have supply-side implications for gas markets. The index has a daily resolution and is normalised so that the average value equals 100 during the base period from January 1985 to December 2019. The graph below depicts the development of the GPRD between 2016-2025, showing among other things a strong increase in geopolitical risk after the Russian invasion of Ukraine in 2022 and the start of the Israel-Hamas war in October 2023.

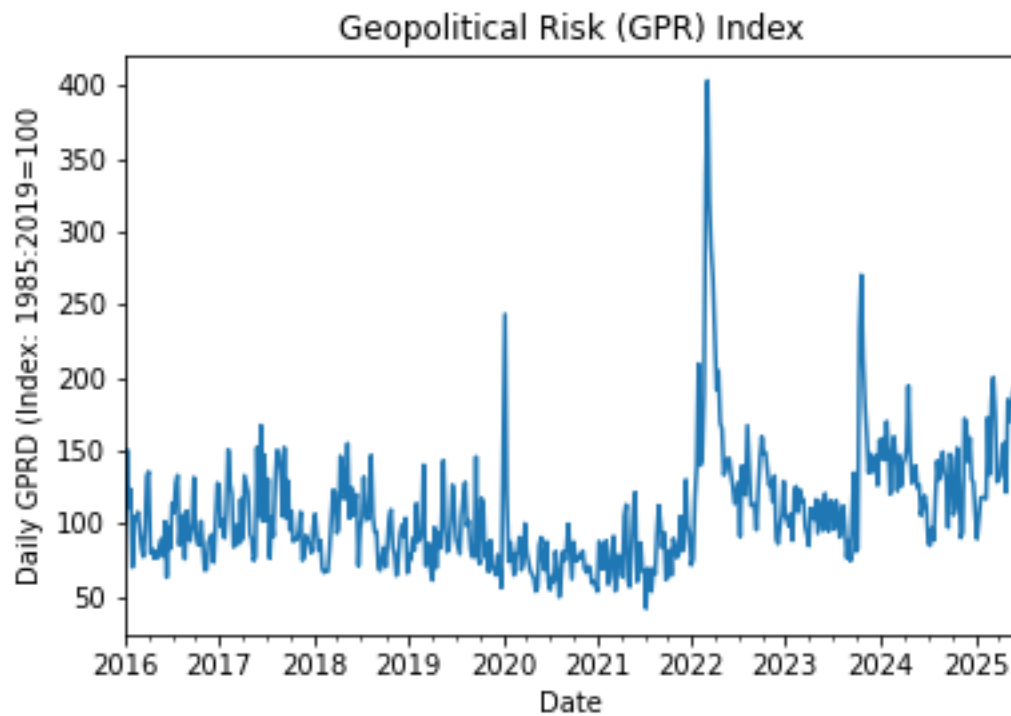


Figure 8: Plot of geopolitical risk indicator indexed to base period 1985-2019

3.3 Data processing

All data variables were downloaded as CSVs and processed in Python using the Pandas library. Variables were merged and rows with NaN values were removed, resulting in a final dataframe with a row for each UK working day from 1 January 2016 – 31 May 2025, comprising 2375 rows and 10 columns before feature engineering:

Variable description	Type	Resolution	Unit
Date	Datetime[ns]	Daily	
TTF M+1 price	Float64	UK working days	€/MWh
Industrial gas demand	Float64	Daily	Mcm
Residential & commercial gas demand	Float64	Daily	Mcm
Gas demand for power	Float64	Daily	Mcm
Storage fullness level	Float64	Daily	%
LNG arrivals to Europe	Int64	Daily	Cargoes
LNG arrivals to Asia	Int64	Daily	Cargoes
LNG arrivals elsewhere	Int64	Daily	Cargoes
Geopolitical risk (GPRD)	Float64	Daily	Unitless

Table 2: Description of each primary variable used in analysis

3.4 Exploratory data analysis

3.4.1 Autocorrelation

Exploratory data analysis is performed to gain a better understanding of the dataset. First it is instructive to inspect the TTF front-month price series. Under the efficient market hypothesis, price levels follow a random walk, meaning today's price is a function of yesterday's price plus some random shock:

$$P_t = P_{t-1} + \varepsilon_t$$

The consequence is that today's price is highly dependent on yesterday's price – this dependence can be measured via autocorrelation which quantifies the linear relationship between a time series and a lagged version of itself. The figure below shows the autocorrelation in the TTF M+1 time series up to 40 lags, demonstrating very high and slowly decaying autocorrelation which is the classic signature of a random walk.

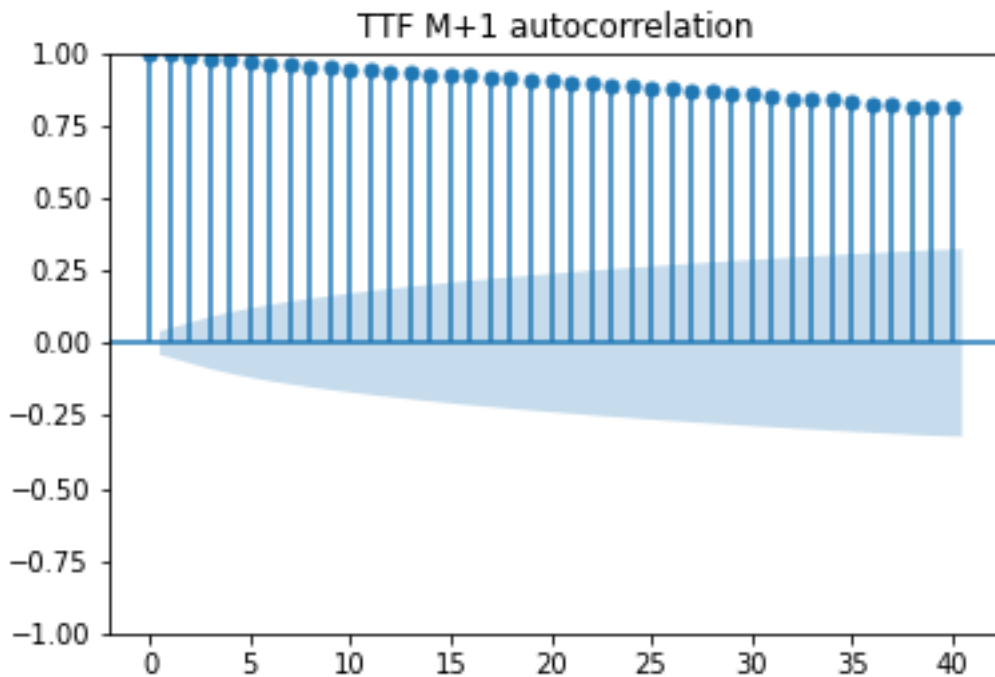


Figure 9: Autocorrelation of TTF M+1 absolute prices

If it is the case that TTF M+1 prices follow a random walk, then daily changes in the price are represented by the error term ε_t , which is random and therefore by definition not correlated with

lagged versions itself. The figure below shows the autocorrelation of daily price changes, which is consistent with the random walk thesis since autocorrelation at all lags except 0 is virtually non-existent. The blue bands represent a 95% confidence interval that the autocorrelation is 0.

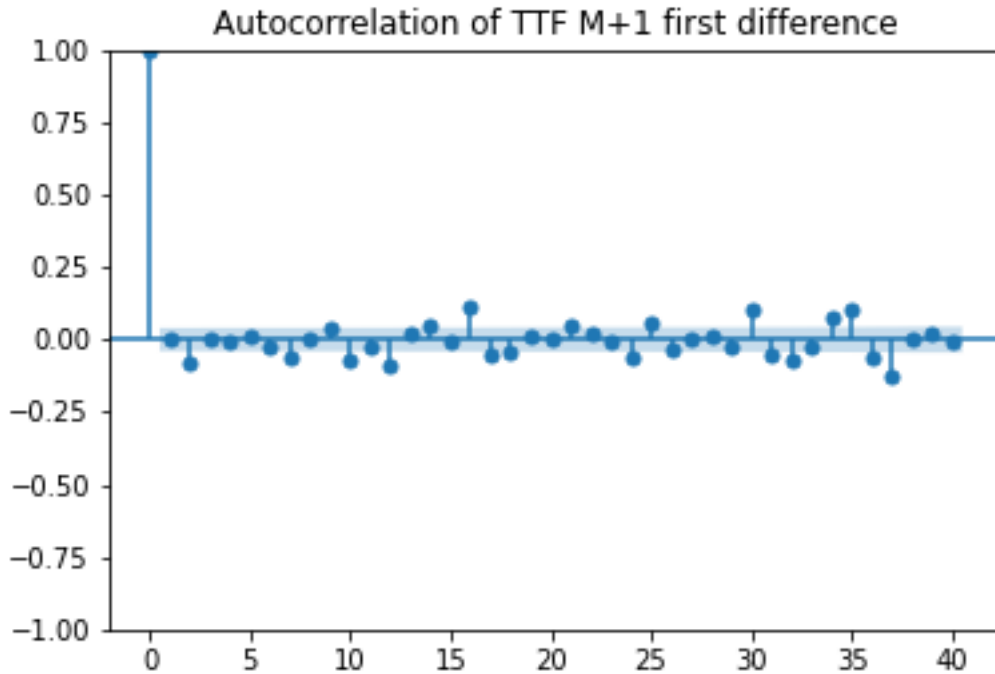


Figure 10: Autocorrelation of daily changes in TTF M+1 prices

3.4.2 Lagged feature correlation

To further explore the explanatory features and inform the feature engineering process, correlation analysis was conducted between the TTF M+1 front-month price and the explanatory variables at positive and negative time lags. This was done to identify which lags exhibit the strongest associations with the front-month price, thereby offering higher explanatory potential.

Spearman's rank correlation coefficient was used rather than Pearson correlation since Spearman is more robust to outliers, can detect non-linear monotonic relationships and doesn't assume normality of the data. The Spearman correlation coefficient is computed as:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where:

- $d_i = \text{rank}(x_i) - \text{rank}(y_i)$

- n is the number of observations

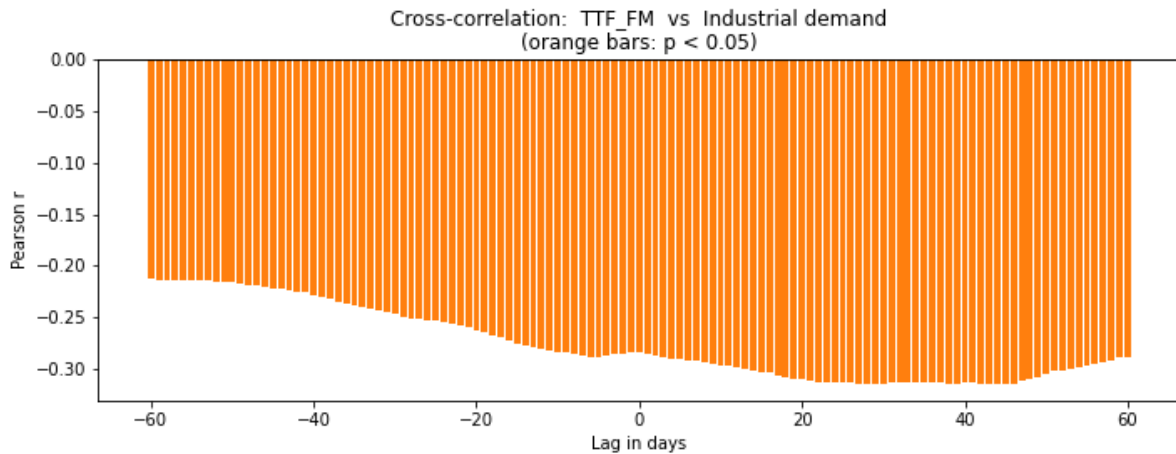


Figure 11: Cross-correlation of TTF M+1 and industry gas demand

The above graph shows that there is a negative correlation between industrial gas demand and TTF M+1 prices over the 2016-2025 period. The negative correlation indicates that when prices are high, industrial demand tends to be low and when prices are low, industrial demand tends to be high. Positive lags represent correlation between the TTF M+1 prices and industrial demand D days earlier, while negative lags correspond to correlation between price and demand D days later.

Correlation at positive lags therefore indicates predictive power of the explanatory variable. In this case, the correlation is strongest at 20-40 positive lags, suggesting lags around this timeframe should be added as explanatory variables.

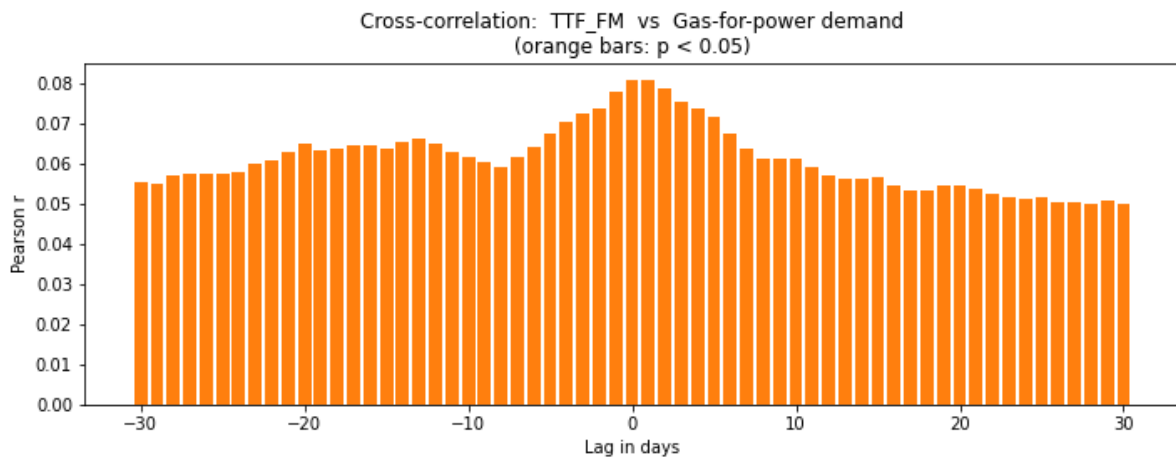


Figure 12: Cross-correlation of TTF M+1 and gas-for-power demand

The correlation between gas demand for power and TTF M+1 prices is weaker, with the strongest correlation at 0 lags. This indicates that including lags is unlikely to improve explanatory or predictive power of the model.

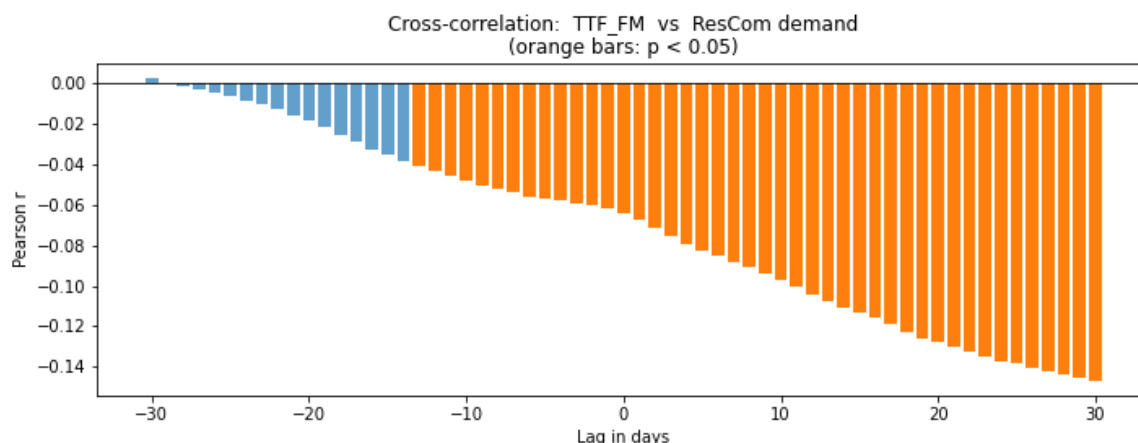


Figure 13: Cross-correlation of TTF M+1 and space heating demand

The correlation between price and Residential & Commercial demand is stronger than for gas-for-power demand but less strong than for industrial demand. The correlation is stronger as the positive lag count increases, suggesting long-term lags for ResCom demand should be included as features.

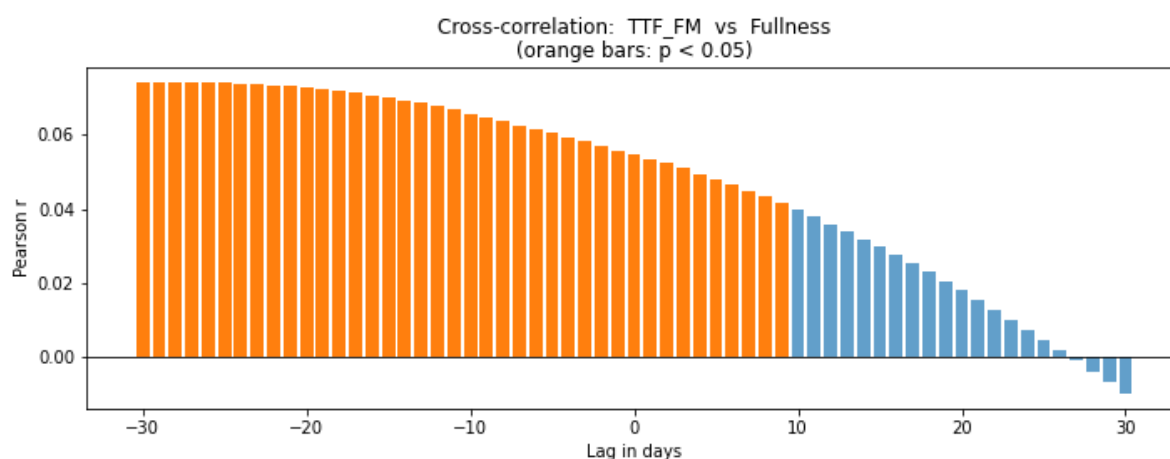


Figure 14: Cross-correlation of TTF M+1 and storage fullness

The correlation between storage fullness levels and TTF M+1 prices across the 2016-2025 period is very weak, and statistically insignificant at positive lags beyond 10, suggesting poor predictive power of storage levels, and that adding storage level lags is unlikely to be useful.

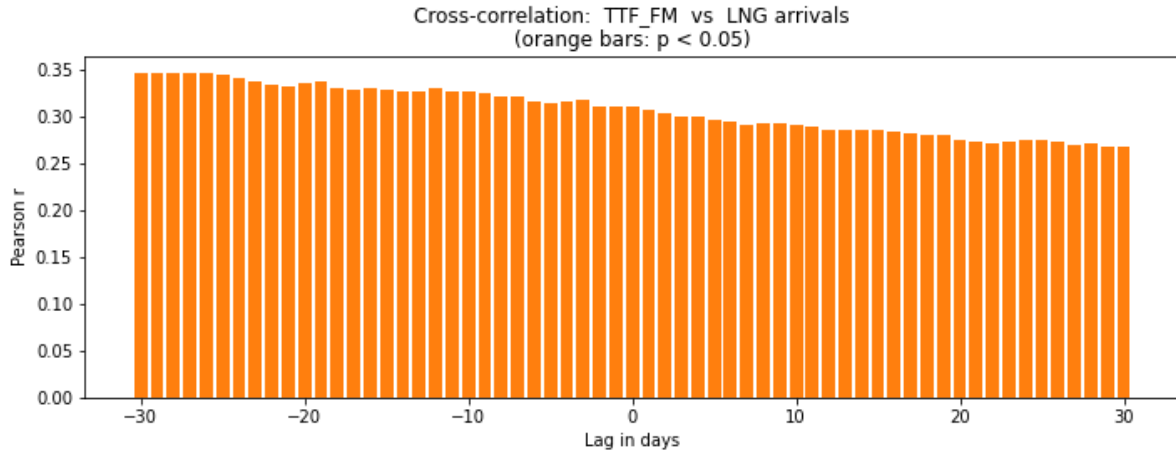


Figure 15: Cross-correlation of TTF M+1 and European LNG arrivals

For LNG arrivals, there is a moderate positive correlation between European LNG arrivals and TTF M+1 prices, meaning that when prices are high, European LNG arrivals also tend to be high. The correlation is stronger for negative lags and decreasing with positive lags, meaning short-term lags such as 3-day, 7-day lags could be more helpful than longer term lags.

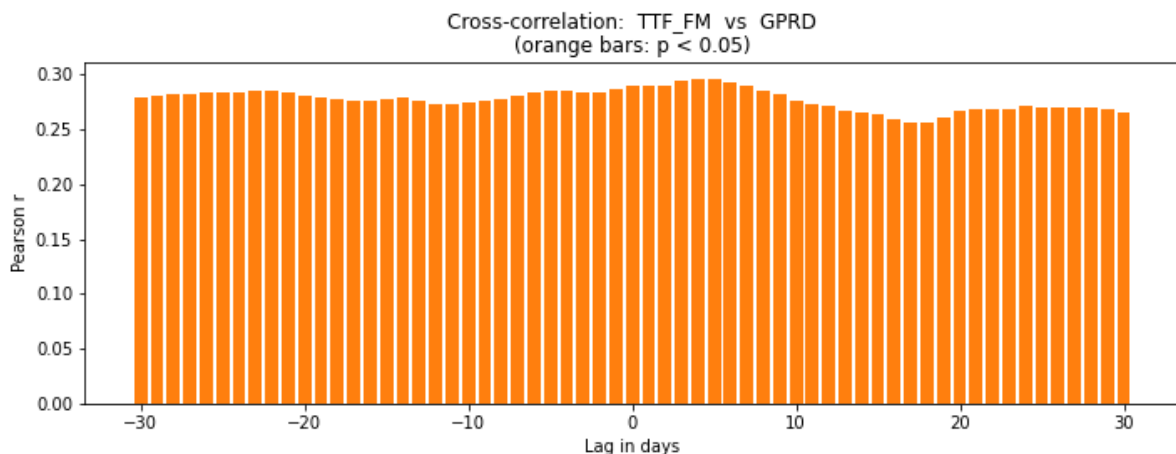


Figure 16: Cross-correlation of TTF M+1 and daily geopolitical risk

Finally, there is moderate positive correlation between TTF M+1 prices and geopolitical risk, as measured by the number of articles related to adverse geopolitical events in major newspapers. The higher geopolitical risk, the higher prices tend to be and vice versa. The correlation is highest near 0-10 positive lags, indicating shorter-term lags will be most useful to include.

3.5 Feature engineering

The first step of the feature engineering process was to take the first difference of the TTF M+1 price series to create a variable 'Diff' which represents the change in price from one working day

to the next. Then, 'Diff' is shifted by 1 row so that each row in the dataset contains the value of the explanatory variables on D-1 and the value of the price change from D-1 to D. The original TTF M+1 absolute price variable is then dropped from the dataset. This structure is the same under both the look-back and look-forward approach.

The remaining feature engineering is unique to each approach. For the look-back dataset, lagged versions of the explanatory features were added as new variables. The lags were chosen based on the correlation analysis performed which revealed a short-term lagged price correlation with geopolitical risk (GPRD) and European LNG arrivals, and longer-term lagged price correlation with residential and commercial demand (ResCom) and industrial demand. The following lags were therefore added:

Variable	3-day moving average	7-day moving average	14-day moving average	30-day moving average
GPRD	x	x		
Europe LNG arrivals	x	x		
ResCom demand			x	x
Industrial demand			x	x

Table 3: Key past lags added under look-back approach

In addition, the 3-day rolling standard deviation, mean and past three days' price changes were included as features, bringing the total up to 23 variables for the look-back approach.

For the look-forward approach, no additional lags were added beyond the original explanatory features which lag one day behind the price change variable 'Diff'. Instead, the following forward-looking moving averages were added:

- Average gas demand for space heating next 14 days: 'ResCom_next_14'
- Average gas demand for power next 14 days: 'Power_next_14'
- Count of European LNG arrivals next 14 days: 'Europe_next_14'
- European share of global LNG arrivals next 14 days: 'Europe_share_14'

This brings the number of variables under the look-forward approach to 14.

3.6 Model architectures

3.6.1 Linear regression

The linear regression model is the simplest model used in this project, serving as a baseline model which the other models can be compared to. It assumes a linear relationship between the input features and the target variable:

$$\hat{y} = \beta_0 + \sum_{i=1}^p \beta_i x_i$$

where:

- $\hat{y}$ is the predicted value,
- x_i are the input features,
- β_i are the learned coefficients,
- β_0 is the intercept.

The model is trained using ordinary least squares (OLS), which minimizes the residual sum of squares:

$$\min_{\beta} \sum_{j=1}^n (y_j - \hat{y}_j)^2$$

Linear regression does not include regularisation by default, meaning it also requires no hyperparameters, and is most effective when the relationship between features and target is approximately linear.

3.6.2 Random forest

The Random forest model is an ensemble of decision trees, where each tree is trained on a bootstrap sample of the data, created by randomly selecting observations from the original dataset with replacement, and a random subset of features at each split. The final prediction is made by averaging the predictions from individual trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

Where T is the number of trees and $f_t(x)$ is the prediction of the tree number t. There are a number of hyperparameter for a random forest model, the most commonly adjusted ones being:

- `n_estimators`: Number of trees in the forest

- `max_depth`: Maximum number of splits from root node to leaf nodes
- `min_samples_split`: Minimum number of samples required to split an internal node
- `max_features`: Number of features to consider when looking at the best split

In general, adjusting each these hyperparameters comes down to an act of balancing model complexity and generalisation, as well as computation time and memory usage.

A key benefit of the random forest model, especially for relatively small datasets such as in this project, is that it includes implicit regularisation through bootstrapping and feature randomness, which helps reduce overfitting.

3.6.3 XGBoost

The XGBoost model is also an ensemble of decision trees, similar to Random Forest, except XGBoost builds trees sequentially, where each new tree corrects the residuals of the previous ones. It optimises the following:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

with:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2$$

where:

- l is the loss function (squared error),
- Ω is the regularisation term,
- w_j are the leaf weights,
- T is the number of leaves,
- γ, λ are regularisation parameters

A key difference between XGBoost and Random Forest is that Random Forest includes implicit overfitting control, whereas regularisation is explicit in the XGBoost model via the Ω regularisation term. In addition to the hyperparameters described above for the Random Forest model, XGBoost includes the following key hyperparameters:

- `learning_rate`: Controls how much each new tree contributes to the overall prediction, acting like a step size in optimisation

- alpha: Adds L1 penalty on leaf weights, like Lasso in regression, driving some leaf weights to zero
- lambda: Adds L2 penalty to leaf weights, shrinking weights and improving generalisation
- early_stopping_rounds: stops training (adding new trees) when the model stops improving on the validation set.

3.6.4 Artificial neural network

A feedforward artificial neural network (ANN) is included to compare ensemble tree models with deep learning approaches. An ANN has the following structure:

- input layer with p features,
- hidden layers with ReLU activation,
- output layer with linear activation for regression.

Mathematically, the ANN with a single hidden layer computes:

$$\hat{y} = W_2 \cdot \sigma(W_1 x + b_1) + b_2$$

where:

- W_1, W_2 are weight matrices
- b_1, b_2 are biases
- $\sigma()$ is the ReLU activation function: $\sigma(z) = \max(0, z)$

Key hyperparameters include:

- hidden_layers: Number of hidden layers (depth of network)
- units_per_layer: Number of neurons in each layer
- activation: Activation function (e.g. relu, sigmoid, tanh)
- optimizer: Algorithm used to update weights (SGD, Adam, RMSprop)
- learning_rate: Step size for weight updates
- batch_size: number of samples per training batch
- dropout_rate: Fraction of neurons randomly dropped during training

ANNs are an extremely customisable model architecture in terms of depth (layers) and width (neurons) per layer, leading to a large amount of parameters which can come at the cost of generalisation, especially in smaller datasets such as the one used in this project.

3.6.5 Long short-term memory

Long short-term memory (LSTM) models are a special kind of recurrent neural networks (RNNs), which are designed for sequential data. When training a neural network, backpropagation is used to update the model weights, where the gradient, a product of partial derivatives of the loss function, shows how much each weight should change in what direction. A problem for sequential RNNs is that the chain of gradients can get very small, “vanishing” so that layers learn very little.

LSTMs solve this vanishing gradient problem by introducing a cell state C_t which makes it easier for the network to retain long-term dependencies. The idea is for the cell state to remember important information for a long time and forget unimportant information. It does this using:

- Cell state C_t - the long-term memory
- Hidden state h_t – the short-term working memory
- Gates – controlling how the information flows through the cell state and hidden state.

At each time step t , given input vector x_t , previous hidden state h_{t-1} and previous cell state c_{t-1} , the LSTM updates are computed as follow:

1. Forget gate f_t – decides how much of the previous cell state to keep:

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f)$$

where:

- f_t is the forget gate vector at time step t
- σ is the sigmoid activation function (outputting values between 0 and 1)
- W_{xf} is the forget gate weight matrix for the current input x_t
- W_{hf} is the forget gate weight matrix for the previous hidden state h_{t-1}
- b_f is the bias vector for the forget gate

2. Input gate i_t – decides how much new information will be added to the cell state:

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i)$$

3. Candidate cell state –represents the new information that could be added to memory:

$$g_t = \tanh(W_{xg}x_t + W_{hg}h_{t-1} + b_g)$$

4. Cell state update – updates the cell state using c_{t-1} , i_t , and g_t :

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t$$

where $\odot$ represents elementwise multiplication

5. Output gate o_t – determines which parts of the cell state contribute to the hidden state:

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o)$$

The key hyperparameters to be tuned in an LSTM are the number of LSTM layers, the number of hidden units per layer, the input sequence length (how many timesteps are fed in at once) and the dropout rate.

3.7 Hyperparameter tuning

Hyperparameter tuning is essential in machine learning because it controls how the model learns, and different settings can provide drastically different results. Suboptimal settings can lead to underfitting or overfitting, while well-chosen parameters improve predictive accuracy and generalization. To ensure consistency, the same tuning procedure was applied across all models and all evaluation periods. For the walk-forward approaches (expanding and sliding window), hyperparameters were tuned only on the first train/test split and then fixed for all subsequent windows. This design was chosen to reduce computational cost while still obtaining representative configurations.

Hyperparameters were optimized using random search. For tree-based models (Random Forest, XGBoost), candidate configurations were evaluated with 3-fold cross-validation. For neural networks (ANN, LSTM), performance was assessed using a 10% validation split of the training data. Random search was chosen over grid search for its efficiency in higher-dimensional spaces and over Bayesian optimisation to avoid additional overhead.

The following subsections describe the model-specific tuning procedures.

3.7.1 Linear regression

Linear regression was used without regularisation (i.e., no Ridge or Lasso penalties) and therefore did not involve hyperparameter tuning. This was intentional, as the linear model served primarily as a baseline to benchmark more complex methods. Ordinary Least Squares (OLS) regression was fitted to the training set without further adjustment.

3.7.2 Random forest

Random Forest hyperparameters were optimised over 25 sampled configurations from the following search space:

- `n_estimators` (number of trees) uniformly from integers in the range 100–999
- `max_depth` (maximum tree depth) from 3–19

- `min_samples_split` (minimum number of samples required to split a node) from 2–9
- `min_samples_leaf` (minimum number of samples required to form a leaf node) from 1–9
- `max_features` (number of features considered at each split) set to one of 'sqrt', 'log2', or None.

3.7.3 XGBoost

Similarly, XGBoost hyperparameters were tuned over sampled 25 configurations from the following search space:

- `n_estimators` (number of boosting trees) uniformly from integers in the range 50–299
- `max_depth` (maximum depth of each tree) uniformly from integers in the range 2–7
- `learning_rate` uniformly from real values in the interval [0.01, 0.21)
- `subsample` (fraction of rows used per boosting round) uniformly from [0.7, 1.0)
- `colsample_bytree` (fraction of features used per tree) uniformly from [0.7, 1.0)

3.7.4 Artificial neural network

The artificial neural network (ANN) was tuned using Keras Tuner's RandomSearch with 10 sampled configurations. The search was deliberately constrained compared with the tree-based models, as training neural networks is significantly more computationally demanding.

The search space was:

- `units_layer1`: {32, 64, 96, 128}
- `units_layer2`: {16, 32, 48, 64}
- `activation`: ReLU (fixed)
- `dropout`: {0.1, 0.2, 0.3, 0.4}
- `learning rate` (Adam optimizer): log-uniform in [0.0005, 0.01]

Each trial was trained for up to 30 epochs with batch size 16, using validation loss (MSE) for tuning. Early stopping with `patience=5` was applied to terminate training once convergence plateaued, reducing overfitting risk.

3.7.5 Long short-term memory

The Long short-term memory (LSTM) network was tuned using Keras Tuner's RandomSearch with 5 sampled configurations. This smaller budget reflects the higher training cost of sequential models relative to both tree ensembles and feedforward ANNs. The search space was:

- units: {32, 64, 96, 128}
- dropout: {0.1, 0.2, 0.3, 0.4, 0.5}
- learning rate (Adam optimizer): log-uniform in [0.0005, 0.01]

Each configuration was trained for up to 20 epochs with batch size 16, using validation loss (MSE) for tuning. Early stopping with patience =5 was applied.

3.8 Feature importance

For models which beat the naïve baseline at a statistically significant level, feature importance was quantified using model-appropriate techniques, described in the following sections.

3.8.1 Linear regression coefficients

For linear regression, feature importance is interpreted directly from the magnitude of the estimated coefficients:

$$\hat{y} = \beta_0 + \sum_{i=1}^p \beta_i x_i$$

where β_i represents the change in the predicted response $\hat{y}$ associated with a one-unit increase in feature x_i , holding all other features constant.

To ensure comparability, all predictors were standardized using z-score normalization prior to model fitting. Thus, the relative magnitudes of the coefficients $|\beta_i|$ can be directly compared as indicators of feature importance.

Coefficient significance was assessed using t-tests, where the null hypothesis $H_0: \beta_i = 0$ was evaluated with:

$$t_i = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}$$

and the associated p-value determined whether a feature contributed significantly beyond chance at 5% significance level.

3.8.2 Ensemble trees

For ensemble tree models, feature importance was derived from split-based impurity reduction. At each split s on feature j , the importance contribution is defined as the reduction in loss:

$$\Delta L_{j,s} = L(\text{parent}) - (L(\text{left child}) + L(\text{right child}))$$

where $L(\cdot)$ denotes the loss function.

In Random forests, the feature importance of variable j is given by the average impurity reduction it provides across all splits and trees:

$$FI_j = \frac{1}{T} \sum_{t=1}^T \sum_{s \in S_{j,t}} \Delta L_{j,s}$$

where T is the number of trees and $S_{j,t}$ the set of splits on feature j in tree t .

In XGBoost, “gain-based” importance was used, which computes the average improvement in the objective function when a feature is used for splitting:

$$FI_j = \frac{1}{|S_j|} \sum_{s \in S_j} \Delta L_{j,s}$$

where S_j is the set of all splits on feature j .

Both methods yield a ranked list of features according to their relative contribution to predictive accuracy. Results were visualised as horizontal bar plots for interpretability.

3.8.3 Neural networks

Unlike linear and tree-based models, neural networks do not attribute importance to features through direct coefficients or splits. Instead, input effects are distributed across multiple layers and nonlinear activations, making interpretability less direct.

To quantify importance, permutation importance was employed. For each feature j , its values were randomly permuted while keeping all others fixed. The resulting change in model error is defined as:

$$PI_j = \frac{1}{M} \sum_{m=1}^M (\text{Err}(f, X^{\pi_{j,m}}) - \text{Err}(f, X))$$

where $\text{Err}(\cdot)$ is the evaluation error, $X^{\pi_{j,m}}$ denotes the dataset with feature j permuted in repetition m and M is the number of permutations.

A higher PI_j indicates that disrupting feature j strongly degrades model performance, implying greater importance.

3.9 Evaluation metrics and statistical significance

The main evaluation metrics reported for each model run in this project are mean absolute error (MAE) and mean squared error (MSE), for the following reasons:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

MAE measures the average magnitude of prediction errors in the same units as the target variable. From a market participant's perspective, this is the most relevant metric: it answers whether the model can get closer to the true ICIS settlement price than simply guessing no change (the naïve baseline).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

MSE punishes larger errors more heavily than MAE, which is important in risk-sensitive contexts where occasional large mistakes may be more costly than frequent small ones.

Machine learning studies often also report R^2 , measuring how much of the dependent variable variance the model explains. However, as discussed in section 3.4.1, the dependent variable in this project represents daily price differences which are approximately white noise, meaning very little variance is explainable. As a result, R^2 values are expected to be near zero and are uninformative in this context.

For models which outperform the naïve baseline at a statistically significant level, two additional metrics are reported for added insight: directional accuracy (DA) and mean absolute scaled error (MASE). DA measures how often the model correctly predicts the direction of change and is defined as:

$$DA = \frac{\text{Number of correct directions}}{\text{Total number of forecasts}}$$

MASE is a scale-independent accuracy measure which directly compares model performance to the naïve baseline:

$$MASE = \frac{MAE_{\text{model,test}}}{MAE_{\text{naive,train}}}$$

If $MASE < 1$, it means the model is outperforming the naïve baseline, for example an MASE of 0.8 means the model is 20% more accurate than the naïve forecast.

To assess whether error improvements over the naïve baseline were genuine or could have occurred by chance, the Wilcoxon signed-rank test was applied to the MAE metric at the 95%

confidence level. Instead of just comparing averages, this test looks at the distribution of paired differences and tests whether these are symmetrically distributed around 0. This checks whether a model is consistently better than the naïve baseline, not just on average.

For each test set observation, the absolute errors of the model and the naïve forecast were computed as:

$$e_i^{\text{model}} = |y_i - \hat{y}_i|$$

$$e_i^{\text{naïve}} = |y_i - 0|$$

The Wilcoxon test was then applied to the paired differences:

$$d_i = e_i^{\text{model}} - e_i^{\text{naïve}}$$

The null and alternative hypotheses were defined as:

$$H_0: \text{Median}(d_i) \geq 0,$$

$$H_1: \text{Median}(d_i) < 0$$

Thus, the test evaluates whether the median paired error is lower for the model than for the naïve baseline. In practical terms, this evaluates whether the model is typically closer to the true ICIS settlement price than guessing no change.

The Wilcoxon test is rarely reported in price forecasting studies but was chosen here for maximum transparency, ensuring that observed improvements were not simply due to random variation.

4. Results and discussion

This section presents the results of applying all five model architectures, along with hyperparameter tuning, to each of the the 6 specified periods and across the expanding window (EW) and sliding window (SW). Results from the look-back approach are presented first – where the model only has access to past information – after which results from the look-forward approach are presented – where the model can peak into the next 14 days of space heating demand, power demand and LNG flows.

4.1 Look-back model performance:

The table below shows the MAE of each model across all evaluation periods and compares it to a naïve baseline of guessing no price change each day. Model-period pairs which perform better at a statistically significant level with 95% confidence level under the Wilcoxon signed-rank test

are highlighted in orange, whereas improvements at non-significant levels are highlighted in green.

	Linear	XGBoost	Random Forest	ANN	LSTM	Naïve baseline
Period 1	0.01864	0.1670	0.1685	0.1676	0.1721	0.1685
Period 2	0.4498	0.4290	0.4135	0.4259	0.3847	0.3969
Period 3	0.2988	0.3091	0.3075	0.3130	0.3067	0.3025
Period 4	1.2351	0.5229	0.5017	0.6268	0.5712	0.5246
Period 5	5.3023	4.8537	4.8335	4.5832	4.6421	4.7037
Period 6	0.9736	0.9274	0.9082	0.9794	0.9600	0.9621
EW	2.0155	1.9200	1.9706	2.3568	2.2754	1.9228
SW	2.2934	1.9650	1.9690	2.5727	2.4487	1.9228

Table 4: MAE of model runs under look-back approach

The results show that all model-period pairs occasionally perform better than the naïve baseline. The best performing model is the XGBoost model which outperforms baseline in 3 out of 6 periods and, notably, in the expanding window which tests across the entire dataset. The XGBoost model is also the only model with a statistically significant result at 95% confidence level, during period 6. The second-best performing models are the Random forest and LSTM models, followed by the ANN and finally the linear model with the worst performance.

Note that the Random Forest model during period 6 has a lower MAE than the XGBoost model despite not being statistically significant. Since the Wilcoxon Signed-Rank test the distribution of paired error differences, this suggests that the Random forest model had a few big outliers/inconsistent gains over the naïve baseline in period 6, whereas the XGBoost model had more consistent improvements.

Evaluating models using MSE reveals exactly the same trend, as shown below.

	Linear	XGBoost	Random Forest	ANN	LSTM	Naïve baseline
Period 1	0.0625	0.0582	0.0584	0.0509	0.0583	0.0596
Period 2	0.2944	0.2542	0.2435	0.2841	0.2333	0.2296
Period 3	0.1344	0.1418	0.1396	0.1401	0.1342	0.1364
Period 4	2.6278	0.4941	0.4810	0.6426	0.5620	0.5107

Period 5	40.9659	36.2767	35.4040	32.0889	34.0783	34.9278
Period 6	1.8156	1.5431	1.5598	1.6374	1.6548	1.5857
EW	29.8174	25.1601	26.6256	42.8752	31.4822	26.1367
SW	35.5267	26.2648	26.4837	39.1501	33.6615	26.1367

Table 5: MSE of model runs under look-back approach

4.2 Look-back model diagnostics and feature importance

As shown above, the XGBoost model performed significantly better than the naïve baseline during period 6, with a Wilcoxon test p-value of 0.0076. This result is significant because it suggests a violation of the efficient market hypothesis during this period – in other words, it would have been possible to consistently “beat the market” using widely available information from the past to predict daily TTF front-month prices.

This period in question spans 9 February 2023 - 31 May 2025, with training on all but the last 60 working days and testing on the last 60. Across the last 60 working days, the model correctly predicted the direction of the next day’s price change 68% of the time. The graph below shows how the XGBoost model consistently had slightly better predictions than a naïve no-change forecast across the test set, leading to a larger gap in cumulative error throughout the test set.

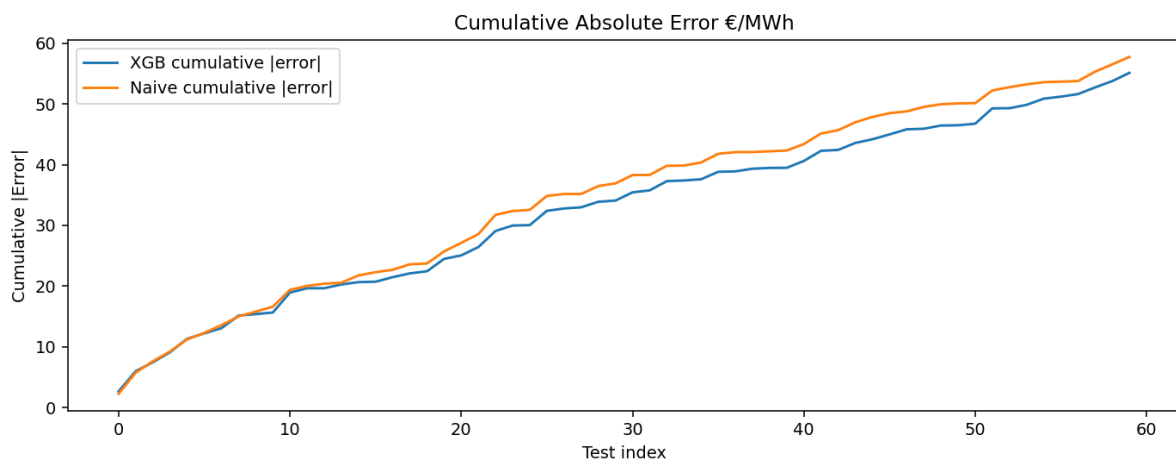


Figure 17: Comparison of XGB period 6 and naïve baseline cumulative MAE

To gain insight into how the XGBoost model made its predictions, feature importance is computed:

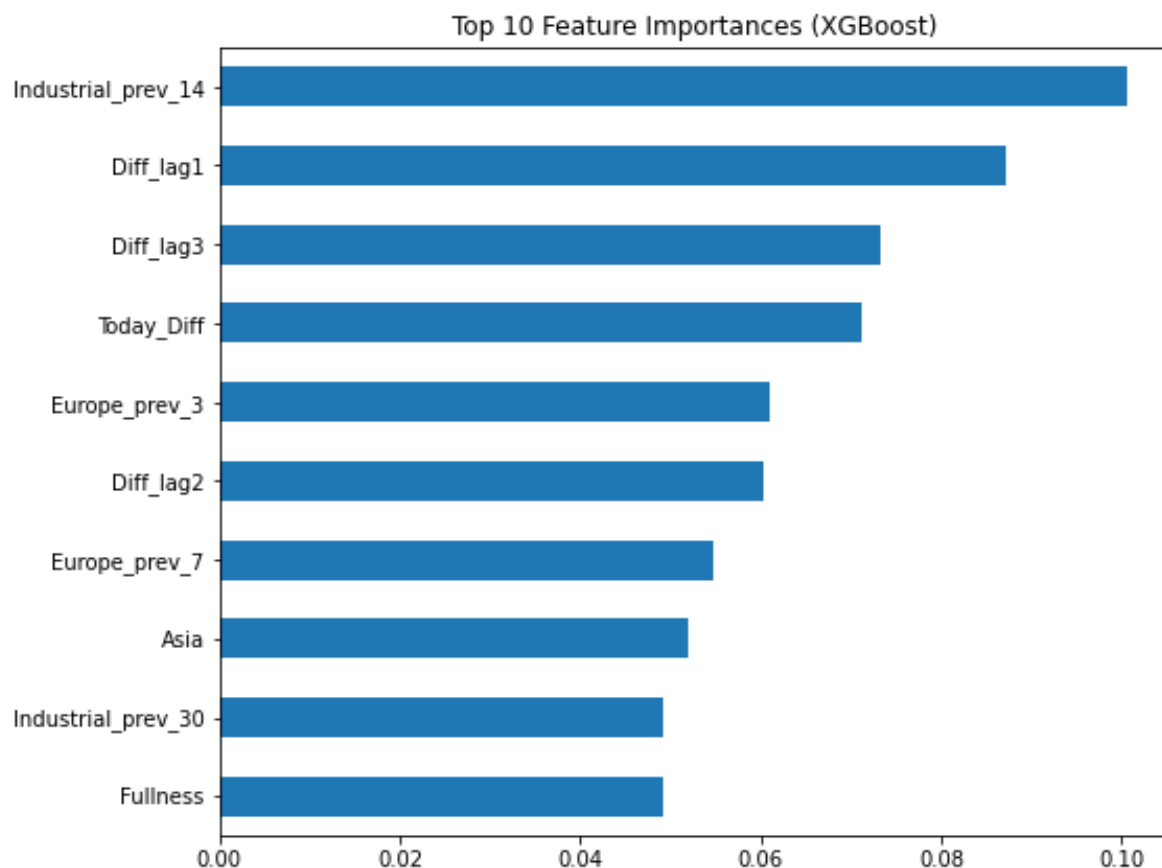


Figure 18: Ranked top 10 XGB period 6 feature importance

The feature importance plot reveals that single most important feature is the 14-day rolling average of European industrial gas demand. This is supported by the correlation analysis performed previously which showed a strong correlation between past European industrial demand and future TTF gas prices.

‘Today_Diff’, ‘Diff_lag1’ and ‘Diff_lag3’ are technical variables rather than fundamental variables, which represent daily price changes prior to tomorrow’s price change. ‘Europe_prev_3’ and ‘Europe_prev_7’ are the sum of LNG cargoes arriving in Europe over the past 3 days and 7 days respectively. ‘Asia’ represents the number of LNG cargoes which arrived in Asia on D-1.

In summary, the variables represented in the top 10 features are: industrial gas demand, technical indicators, LNG cargo arrivals and storage fullness. It is interesting to note that no variables related to geopolitics are represented in the top 10 features – this could mean either that past information on geopolitical tensions are not useful for predicting gas prices in general, or just that they were not useful specifically in the post-energy-crisis period between February 2023 – June 2025.

4.3. Look-forward model performance:

The below table presents the MAE of each model using the look-forward approach and compares to the naïve baseline. Under the look-forward approach, the model gets access to both backward-looking variables and four additional variables which peak into the future of LNG flows and weather-driven demand. Green cells are improvements over naïve baseline, while orange cells show improvements at a statistically significant level under Wilcoxon signed-rank test.

	Linear	XGBoost	Random Forest	ANN	LSTM	Naïve baseline
Period 1	0.1926	0.2074	0.1710	0.1771	0.1775	0.1685
Period 2	0.3765	0.3694	0.3797	0.4048	0.3930	0.3969
Period 3	0.3005	0.2979	0.2918	0.3567	0.3417	0.3025
Period 4	0.6904	0.5278	0.5135	0.5535	0.6009	0.5246
Period 5	5.5047	4.6266	4.5713	4.6216	4.6008	4.7037
Period 6	0.9854	1.0558	1.0540	1.0120	1.0732	1.0796
EW	1.9334	1.9791	1.9318	2.0726	1.9511	1.9172
SW	2.1459	1.9993	1.9637	2.4591	2.2277	1.9172

Table 6: MAE of model runs under look-forward approach

The results show that the Random forest model outperformed the naïve baseline in each of the 6 individual periods, but not at a statistically significant level. On the other hand, the XGBoost model significantly outperformed the naïve baseline during period 2, and so did the LSTM model during period 5 – despite having a higher MAE than the Random forest model. Highly surprisingly, the linear model is the only model to outperform the naïve baseline at a statistically significant level during period 6. No models were able to beat the naïve baseline consistently in either of the walk-forward validation approaches.

Evaluating the period-model pairs with MSE reveals the same results:

	Linear	XGBoost	Random Forest	ANN	LSTM	Naïve baseline
Period 1	0.0651	0.0777	0.0574	0.0610	0.0626	0.0596
Period 2	0.2196	0.2073	0.2152	0.2428	0.2270	0.2296
Period 3	0.1350	0.1432	0.1299	0.1993	0.1850	0.1364

Period 4	0.8401	0.5073	0.4900	0.5783	0.6247	0.5107
Period 5	38.0636	33.3186	32.6435	33.5678	33.5860	34.9278
Period 6	1.8865	1.9889	1.9745	1.8730	2.0482	2.0717
EW	26.6111	26.8029	26.8356	29.9437	26.3465	26.1271
SW	30.0600	27.7642	27.0845	35.4851	31.7293	26.1271

Table 7: MSE of model runs under look-forward approach

4.4 Look-forward model diagnostics and feature importance

Three look-forward model-pairs performed better than the naïve benchmark at a statistically significant level: the XGBoost model during period 2, the LSTM model during period 5 and the linear model during period 6.

The XGBoost model, applied to the period 2 November 2017 – 14 March 2019, significantly outperformed the naïve baseline with a p-value of 0.0113, directional accuracy of 58.3% and MASE of 0.775. This allowed it to make consistently better predictions than a naïve baseline across the 60-row test set, building up lower cumulative error.

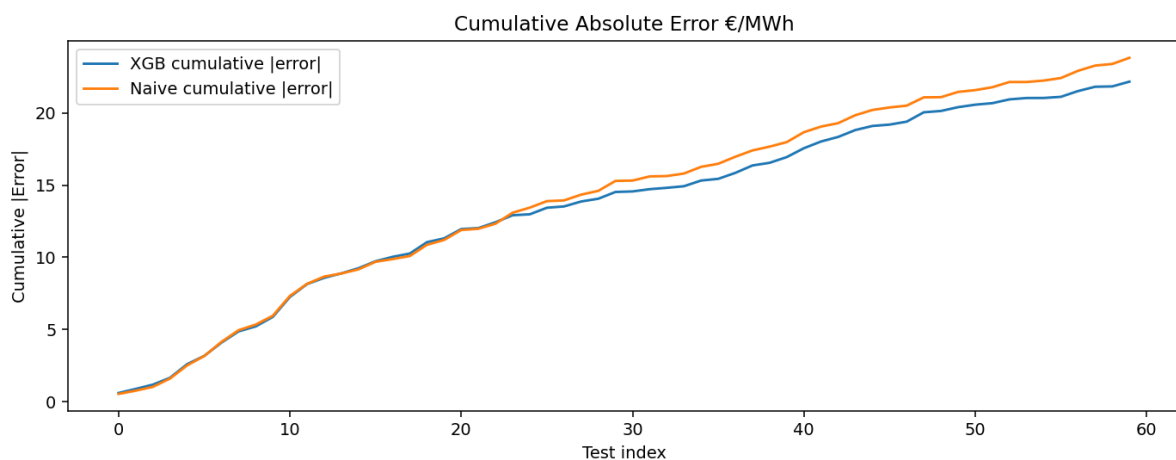


Figure 19: Comparison of XGB period 2 and naïve baseline cumulative test MAE

The graph below shows the top 10 feature importances used by the model to make consistently better predictions.

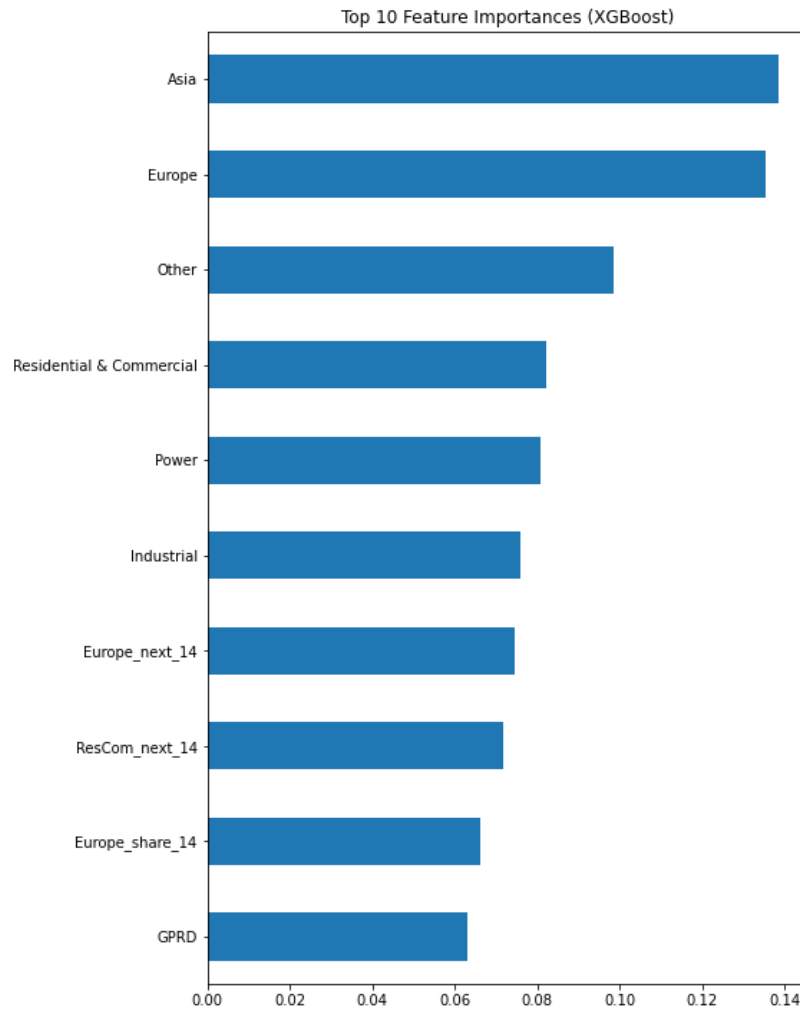


Figure 20: Ranked top ten XGB Period 2 feature importance

Surprisingly, although the forward-looking variables appear in the top 10 list over most important features, they are not in the top 5. Rather, the three most important variables are ‘Asia’, ‘Europe’ and ‘Other’, all of which are backward-looking and refer to yesterday’s LNG arrivals in Asia, Europe and elsewhere.

The strong performance of the XGBoost model during period 2 may therefore not be primarily due to the addition of forward-looking variables but rather due to the simpler set of explanatory features used, with only D-1 features, and a few forward-looking features.

The second statistically significant result was the LSTM model during period 5 – from 3 June 2021 to 10 February 2023, the peak of the energy crisis. This model had a p-value of 0.046, just within the significance threshold, a directional accuracy of 60% and an MASE of 0.656. This enabled it to build up lower cumulative error than the naïve baseline, particularly toward the end of the test set, as shown below.

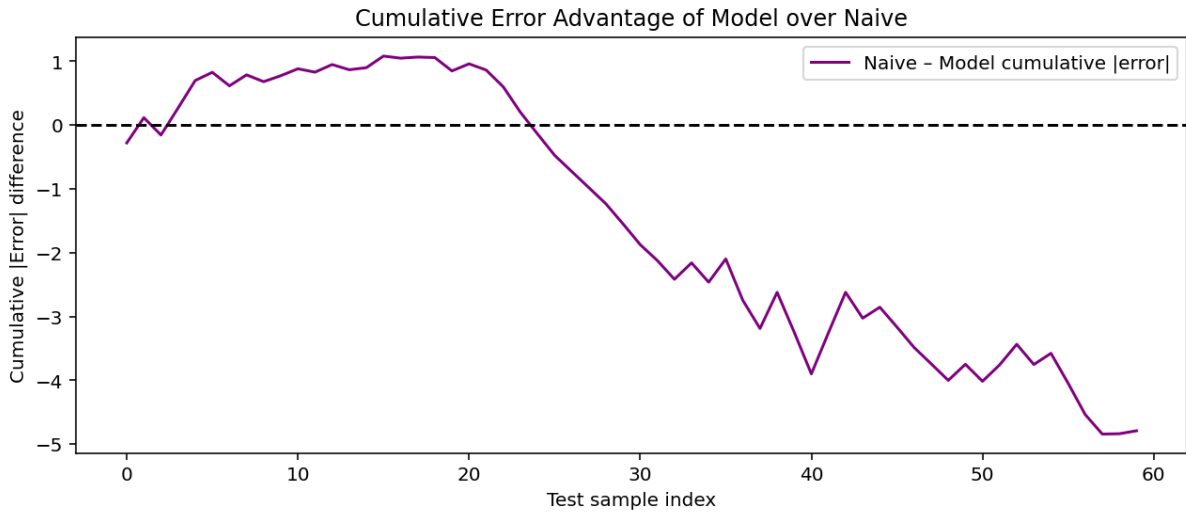


Figure 21: Cumulative difference in MAE between naïve and LSTM model

To investigate which variables were particularly important for predictions during this period, the graph below displays permutation-based feature importance for the model:

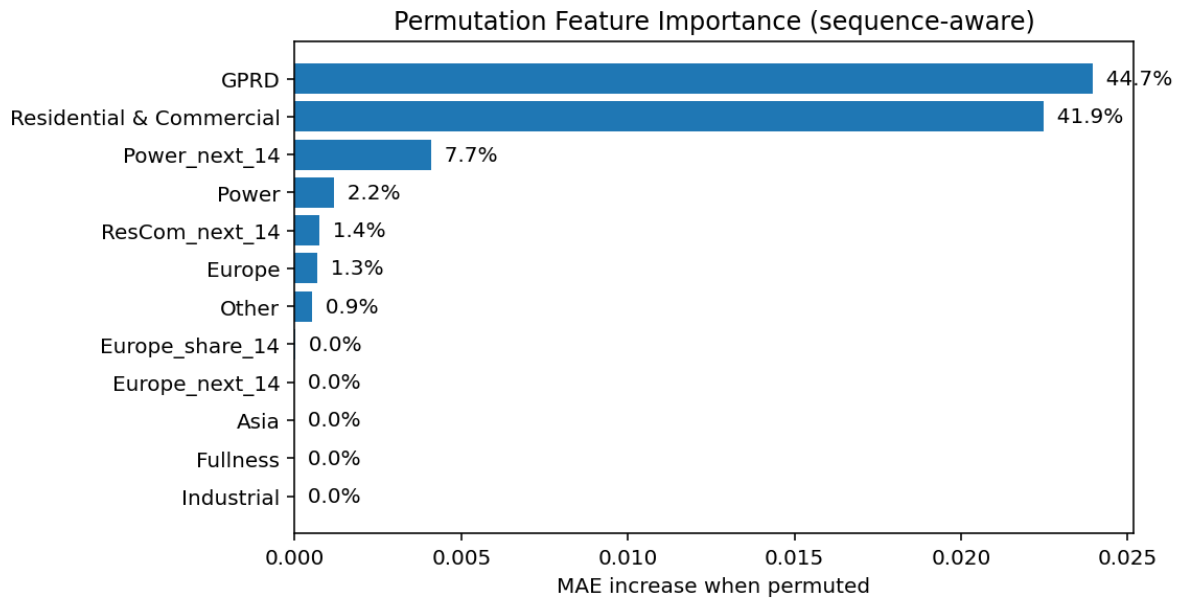


Figure 22: Permutation based feature importance for LSTM period 5

This feature importance chart shows a strong contrast to the feature importance during period 2. This time, variables related to LNG arrivals were largely irrelevant whereas the daily geopolitical risk indicator had a high degree of importance. This is interesting since the GPRD value the model has access to is the value from D-1, implying that during this period the market partly reacted to geopolitical news headlines with a one-day lag. In other words, simply

knowing the news from today could give market participants an edge in predicting tomorrow's gas price.

The third statistically significant result was the linear regression model during period 6 – from 10 February 2023 to 31 May 2025. The model had a directional accuracy of 68% and an MASE of 0.745. This allowed steady accumulation of less error than the naïve baseline, as shown below.

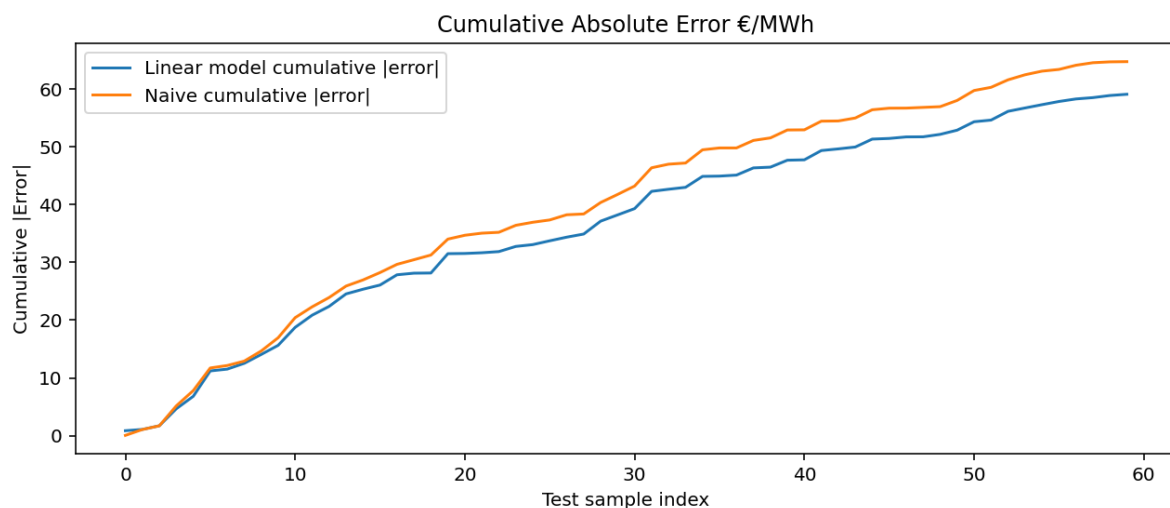


Figure 23: Cumulative error of linear model against baseline naïve

To investigate feature importance, the model coefficients can simply be examined.

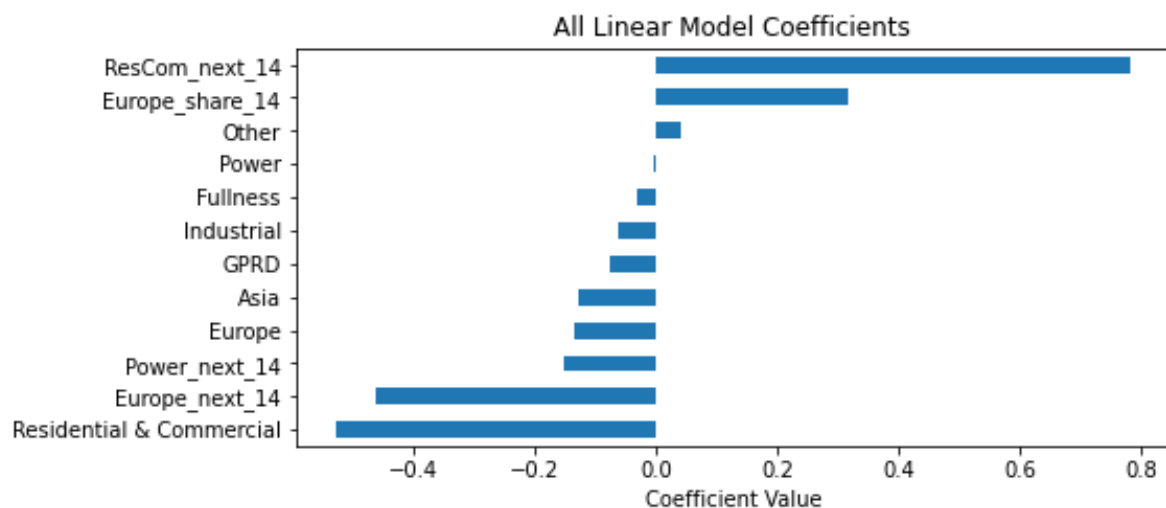


Figure 24: Overview of linear model coefficients

Since features have been scaled, coefficients cannot be interpreted as directly as if the features had been unscaled. The coefficients here instead represent the effect of a 1-standard-deviation

change in the feature on the predicted target, which is tomorrow's day-on-day change in the TTF front-month contract (in €/MWh).

The coefficients reveal that the average space heating demand in the residential and commercial sector over the next 14 days is the single most important feature for the model, with high future space heating demand being associated with higher prices. Europe_share_14, which denotes Europe's share of global LNG arrivals over the next 14 days, also had a high positive coefficient. This indicates that a higher European share of LNG arrivals is associated with higher prices, likely as it represents European willingness to buy more cargoes to cover demand.

Among negative coefficients are D-1 Residential & Commercial space heating demand, LNG cargo arrivals over the next 14 days and gas for power demand over the next 14 days. The remaining variables have very low coefficient values.

In fact, t-tests from the OLS model reveals that ResCom_next_14 is the only statistically significant coefficient at the 5% significance level. This suggests that if one had a very good 14-day forecast for space heating demand during period 6, a simple linear model with this variable would be able to "beat the market".

5. Conclusion, limitations and future work

A range of conclusions can be drawn from this work. First, no model was able to outperform the naïve last-observed-value predictor at a statistically significant level across the entire dataset – consistent with the hypothesis that the TTF is an informationally efficient market. However, during certain periods, certain models were able to beat naïve forecasts at a statistically significant level, violating a strict form of the efficient market hypothesis.

The XGBoost model emerged as the best-performing model, while the Random forest and LSTM models were strong contenders. The ANN model performed poorly under the both the look-back and look-forward approach, while the linear model drastically improved performance when it had access to forward-looking variables. This demonstrates that with the right variables, such as accurate demand forecasts, even simple linear models can beat the market and outperform more complex models. The results also supports the literature that finds ensemble tree methods outperform deep learning approaches on medium-sized tabular datasets.

The second key insight from this research is that gas price drivers vary by period, with the following drivers per period identified:

- Between 2 November 2017 – 14 March 2019, variables related to LNG arrivals in Europe, Asia and elsewhere were particularly important for XGB model predictions
- Between 3 June 2021 - 10 February 2023, D-1 news articles related to geopolitical events gave the LSTM model an edge in predicting prices
- Between 10 February 2023 - 31 May 2025, the past 14 days of industrial gas demand and future 14 days of space heating demand emerged as particularly important factors for the XGB and linear models

The change in importance of fundamental factors over time is likely the reason why, although no models consistently beat the naïve baseline at a statistically significant level throughout the dataset, certain models could do so during certain periods.

There are a number of limitations to the presented work, each of which suggest improvements for future research. Firstly, the number of dataset observations were a key limitation of the project - since ICIS sector-level gas demand data was only available from 2016 onwards, and since ICIS settlement prices are only assessed once per UK working day, the number of total rows in the dataset was only 2375. This makes it difficult to train complex machine learning models without significant overfitting. Future work could mitigate this limitation by using high-frequency trade data rather than daily settlement prices, or obtain variables available over a longer time period, to more robustly test the performance of deep learning approaches.

In a higher-volume dataset, a wider variety of features could also be used, such as other commodity prices, macroeconomic indicators and physical gas flow data, which could give a richer interpretation of important features. Furthermore, this research relied on feature selection based on domain knowledge and literature emphasis, and feature engineering based on correlation analysis. Future research with a richer dataset could explore more formal feature selection algorithms, which systematically filter out the most relevant features for the final models.

Another key limitation for the project was the limited exploration of hyperparameter tuning spaces, especially for the deep learning models, due to computational constraints and the need to run a large number of models. Future research can perform more extensive hyperparameter searches to see whether this will meaningfully improve the performance of the deep learning models.

Finally, analysis of feature importance was prioritised in this research over more detailed model diagnostics, aiming to provide insight on TTF market behaviour. Future research, particularly if

supplemented by richer data and more comprehensive hyperparameter tuning, could focus on the model diagnostics side, investigating more thoroughly why the models behave the way they do.

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Appendix

Step A6 – MSC ESDA Dissertation Ethics Declaration



Statement of Risk Assessment & Ethics Approval Requirements
Student Candidate Number: LSHM1 Student Name: [REDACTED] Student UCL Email Address: [REDACTED] Supervisor Name: Dr Amir Gharavi Supervisor UCL Email Address: a.gharavi@ucl.ac.uk Dissertation Research Proposal [FILL IN]: - Title / Topic: Machine learning for gas price forecasting - Research Question(s) / Aims & Objectives: - Data & source (specify all data to be used; if none, explain why): Energy-specific variables including gas demand, storage levels, LNG flows and prices from ICIS, as well as a geopolitical risk indicator from Dario and Iacoviello (2022) - Method(s) (specify all methods to be used): Machine learning models will be run in Python and performance compared against a naïve no-change benchmark
I have read and understood Step A1 'Does the research require a Risk Assessment?' and: This planned research does NOT require a risk assessment.
I have read and understood Step A2 'Does the research require External research ethics approval?' and: This planned research does NOT require external ethics review.
External ethics approval is <i>not required</i> and I have read and understood Step A3 'Is the research Exempt from the need for ethics approval?' and: This planned research IS EXEMPT from the need for research ethics approval.
The research is <i>not exempt</i> from the need for ethics approval and I have read and understood Step A4 'Does the research require High Risk ethics approval?' and: [DELETE ONE STATEMENT]: This planned research IS deemed high risk and approval from the UCL Research Ethics Committee will be secured before the research starts. This planned research is NOT deemed high risk.
The research is <i>not exempt</i> from the need for ethics approval, does not require high risk ethics approval and: I have read and understood Step A5 'Does the research require ESDA low risk ethics review for questions-based methods OR BSEER low risk ethics review for other methods?' and: [DELETE ONE STATEMENT]: This planned research requires MSc ESDA Low Risk Ethics approval for questions-based methods and approval will be secured before data collection starts. This planned research requires BSEER low risk ethics approval (for other methods), which will be secured before data collection starts.
I confirm that: <u>the</u> information I have provided is accurate to the best of my knowledge. <u>if</u> the answers to any of these questions <u>changes</u>, I will go through this protocol again.

NEXT STEPS:

- STUDENT: Copy the text of the *completed* statement above into an email and email it to your supervisor.
- SUPERVISOR: Reply to the email confirming your approval of the completed statement, copying the Dissertation PGTA (r.alasmar@ucl.ac.uk). It is the student's responsibility to ensure that happens.
- STUDENT:
 - Include this A6 Statement as a Dissertation Appendix after you have BLACKED OUT YOUR NAME & EMAIL ADDRESS so the second marker can mark anonymously.

The Dissertation mark sheet asks the second marker whether this form was filled out correctly and, if not, what % mark deduction they recommend.